

**BACK TO  
METAL**

WHY THIS BOOK EXISTS

# YOU ARE ALLOWED TO RUN YOUR OWN COMPUTERS

A generation of engineers has now been trained to believe that owning a server is a kind of professional failure. It is not. It is a trade, and like every trade it has terms: you choose dedicated capacity, rented or owned in colocation, and keep more of the budget as your workload grows. The cloud-ops team transitions to on-prem ops with the same planned hours, while remote hands handles physical interventions. For a company with steady demand and a substantial cloud bill, both routes deserve a serious comparison.

What has been missing is not the argument. It is a plan short enough to finish. Repatriation is written about as a decision and executed as a crisis, which is why so many attempts stall halfway, with two platforms live and nobody able to say whether it is going well. This is the other thing: 20 Moves, in five stages, each one a single job with a stated cutover, a stated risk and a rollback that works. An earlier edition of this book had a hundred and twenty-two of them and took four years. Nobody has four years.

It is honest about the parts that do not pay. One whole Move ends by telling you to keep paying somebody else, because a content network, denial-of-service scrubbing and outbound email are three things you will not beat on your own, and a book that pretends otherwise is selling something. Another gives you permission to read three Moves, do the arithmetic, and stop. The point was never to own everything. It was to own the parts where ownership is cheaper, faster and yours.

Which is why the whole thing is **open source**. Every Move, the typesetter that turns them into this book, and the site that publishes them are yours: the text under Creative Commons, the software under the MIT licence, all of it at [github.com/OneUptime/back-to-metal](https://github.com/OneUptime/back-to-metal). Correct it, extend it, add the services your own estate actually runs. Infrastructure knowledge this practical should not sit behind a consulting invoice.

# BACK TO METAL

How a company leaves the cloud, in twenty moves

**Nawaz Dhanda**

BY THE MAKERS OF ONEUPTIME • [ONEUPTIME.COM](https://oneuptime.com)

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## BACK TO METAL

How a company leaves the cloud, in twenty moves

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Published by HackerBay, HackerBay.io

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First edition, 2026. Version 6.4.0.

Every runbook in this book changes production infrastructure, and several of them move data that cannot be recreated. Read the Rollback section before the first step of any Move, take a backup you have restored at least once, and rehearse against a copy. Prices, service names and product behaviour are those the author observed while writing and will drift; the figures in each Move are illustrative of the shape of a saving, not a quotation. Nothing here is legal, financial, security or compliance advice. You remain responsible for your own systems, your own data and your own obligations to the people whose data it is.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Move would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are licensed under the SIL Open Font License.

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# MANY MOVES, NOT ONE MIGRATION

The reason leaving the cloud fails is almost never that the technology did not work. It is that it was attempted as one decision, executed as one project, and abandoned somewhere in the middle with two platforms running and nobody able to say whether it was going well.

This book takes the opposite shape. It is 20 Moves, in five stages. Each is one job with a stated cutover, a stated risk and a rollback that has been thought about. Each can be done on a Tuesday and undone on a Wednesday. You can stop after any of them and still be in a coherent place, which is the only property that matters in a project measured in quarters.

It is deliberately small. The first edition had a hundred and twenty-two Moves and, computed from its own files, 975 person-days and four and a half years. That was correct and useless. This is 107 person-days and about 44 weeks for two people, most of it hardware lead time rather than work. Hiring does not shorten it.

They are ordered so that a reader who starts at Move 01 and works forward has, by construction, met every prerequisite of the Move in front of them. That is not a stylistic choice; the build refuses to compile a book where it does not hold.

Every Move carries all three clouds. The job is the same whether you are leaving AWS, Google Cloud or Azure — what differs is the extraction, so each Move opens with three lines naming the real service on each provider and the one thing that is different there.

Move 04 concludes that you should keep paying somebody else for three things. A content delivery network, scrubbing capacity at the edge and outbound email are businesses other people run better than you will, and each is cheap next to what it replaces. The aim was never to own everything. And Move 03 gives you permission to stop: if the arithmetic does not clear a third, three Moves and a week is the whole cost of finding out.

**Rehearse everything, and read the Rollback before the first step of every Move — including the ones that look like nothing.**



EVERY MOVE, WITHOUT EXCEPTION

## HOW A MOVE WORKS

- 1 THE RING**  
The cutover: how many minutes of user-visible downtime this Move costs, against a 60-minute dial. An empty ring means none, and most rings are empty.
- 2 THE BARS**  
Risk, meaning blast radius if it goes wrong rather than how fiddly it is. Three bars and you do not run it alone.
- 3 THE PANEL**  
Everything that must already be true before step one. Hardware, software with versions, and the access you will be asked for.
- 4 WHY THIS WORKS**  
What the managed service actually was, and why the replacement holds. Read it once; you will not need it again.
- 5 THE STRIP**  
What it cost, what it costs now, the saving, the downtime and the person-days. Illustrative of a shape, not a quotation for your account.
- 6 THE ROLLBACK**  
On the facing page, at the foot, in colour. It names the point of no return. Read it before step one, not after step five.

# WHY LEAVE AT ALL

Colocation and rented dedicated servers put more of your infrastructure budget into capacity you can use. Rent the machines to preserve cash, or own them in a colocation facility for lower running costs and control over the hardware. In this reference

model, colocation saves \$4,886,320 over five years, with the same operations hours as the cloud. Your team brings its skills; the provider supplies the building and physical support.

WE DID THIS, AND HERE IS WHAT ACTUALLY HAPPENED

## 730 days

at 99.993 per cent measured availability, through a period that included a region-wide outage at the cloud we had left

## 19%

lower latency on the same software, from local NVMe and the absence of neighbours. We bought no faster code to get it

## 2

hardware interventions in twenty-four months, both disks, both handled by remote hands with a mean response of twenty-seven minutes

## 0

people hired. The toil moved; it did not multiply. We measured it at about fourteen engineer-hours a month across two sites

Our own fleet and our own measurement, not an independent audit, and a bigger estate than this book is sized for. The rest of the figures here stand on outside sources and on arithmetic you can repeat.

### 1 THE BILL STOPS BEING A PERCENTAGE OF YOUR GROWTH

This is the one that matters and the rest are consequences of it. A cloud bill is a toll on activity: more users, more bytes, more bill, for ever, at a margin somebody else sets and can change. A rack or a dedicated-server contract buys capacity at a predictable price. While your workload fits that capacity and your power and traffic allowances, more users need not mean more infrastructure rent. Add machines when demand calls for them. Over five years the difference here is \$4,886,320, and the machines are still working at the end of it.

#### CHECK IT

Take last quarter's invoices and plot the total against your own usage metric. If the line is flat you have nothing to gain here. If it tracks your growth, that slope is what you are buying out of — Move 01.

### 2 EGRESS STOPS BEING A TAX ON YOUR OWN TRAFFIC

Moving your own data to your own users is where the margin lives, and it is the line that ends most of these arguments: \$7,980 a month to send a hundred terabytes out. Transit at a facility is bought by the megabit and costs a small fraction of that. Nothing you build gets faster and no code changes; the line simply stops being there.

#### CHECK IT

Find the data-transfer line on last month's bill. Divide it by your egress in terabytes. If it is anywhere near ninety dollars a terabyte, that line is nearly pure margin — Move 01, then Move 18.

### 3 THE MACHINES ARE YOURS, SO THE PERFORMANCE IS YOURS

A vCPU is half a core somebody else is also using. Local NVMe is not a network service with a queue in front of it and a token bucket on top. The gain is not theoretical and it is not small: on identical software we measured nineteen per cent, and the noisy-neighbour tail — the ninety-ninth percentile that wakes people — improves more than the mean does.

#### CHECK IT

Compare your ninety-ninth percentile against your median for a week. If the gap is wide and unexplained by your own code, some of it is not yours — Move 14 measures it on real hardware before anything moves.

### 4 NOBODY DEPRECATES YOUR HARDWARE

A managed service is somebody else's roadmap running inside your product. Instance families are retired, versions go end-of-life on a date you did not pick, a control plane upgrades on its own release channel, and the longest you can hold it is ninety days. A machine you own runs the version you chose until you choose another one, and the migration happens when you have time rather than when the notice arrives.

#### CHECK IT

Count the forced upgrades and deprecation notices you have absorbed in the last two years, and what each cost in engineer-days. That is a recurring bill nobody invoices you for.

### 5 YOUR TEAM RUNS THE PLATFORM; REMOTE HANDS LOOK AFTER THE RACK

Your cloud-ops engineers already deploy, monitor, patch and recover services. Those skills move with the workload. Colocation remote hands carry out disk swaps, cabling and power cycles under your runbooks; a rented-metal provider maintains its hardware under the support agreement. The reference model keeps the same team and monthly operations hours across all three options.

#### CHECK IT

Agree the physical tasks, coverage and response times with the provider, then rehearse an incident together. Measure your team's hours before and after the move — Moves 07 and 19.

# CHOOSE HOW YOU MOVE

Both routes can lower the bill. Choose the purchasing and support arrangements that fit your

cash flow and the team you already have.

## 1 RENT FIRST, OR OWN IN COLOCATION

Rented metal lets you move without buying the servers upfront. Colocation lets you own the hardware and spread its purchase cost over years of use. Compare both cash flows in Move 03 and choose the route in Move 07.

## 2 A TIMETABLE THAT FITS THE ROUTE

Available rented servers can avoid the hardware purchase and facility setup lead times. The colocation plan includes ordering machines, signing space and waiting for circuits: 107 person-days of work across 44 weeks. Both routes keep the migration rehearsals and rollback windows.

## 3 THE SAME TEAM, THE SAME OPERATIONS HOURS

Cloud ops transitions into on-prem ops. Remote hands handles the physical interventions; your engineers keep responsibility for the platform and application. The model budgets 160 hours a month in every option, with remote hands in the facility bill. Migration work is counted separately.

## 4 THREE THINGS THAT NEVER COME HOME

A content delivery network, outbound mail deliverability and scrubbing at the edge are businesses other people run better than you will. Move 04 tells you to keep paying for all three. These and the services retained in later Moves total \$9,180 a month in the reference model.

## WHEN TO STAY EXACTLY WHERE YOU ARE

Your load is spiky or seasonal and genuinely scales to near nothing between the spikes. Elasticity is the one thing a rack cannot do, and paying for the peak all month is how owning becomes the expensive answer.

You have no team to own the platform and no managed operations partner. Remote hands covers physical work; application recovery and platform decisions still need an owner. An existing cloud-ops team can transition into that role.

You lean hard on managed services that have no real equivalent - a serverless database, a hosted stream, a workflow engine - and replacing them is a rewrite rather than a migration. Move 02 will tell you this in an afternoon.

Your effective cloud bill is too small to cover migration and the capacity you need. Price rented metal as well as a colocation rack: renting can suit a smaller estate. Move 03 compares your own quotes and hours, so the decision rests on your workload rather than a spending threshold borrowed from another company.

If none of those four is true and the arithmetic in Move 03 clears, the rest of this book is twenty jobs that get you there, each one done on a Tuesday and undone on a Wednesday.

# THE REFERENCE BUILD

Every Move in this book is written against one cluster, so that a runbook can name a real thing rather than a category: one site, 16 nodes and 2 on the shelf, 32 cores and 256 GB a node, 25 GbE to the switch. 16 because that is what the bill buys and a little over, and because at this size losing one costs six per cent of the fleet rather than a third; the spares because a dead board is a return authorisation and three weeks. Scale the numbers; do not scale away the redundancy.

Cloud VMs move to KVM guests on Proxmox VE hosts; several guests share each physical machine. Already-containerised workloads use Talos and Kubernetes, directly on hardware or in guests when sharing a VM estate. Classify the workloads in Move 02 before assigning hosts in Move 11. These figures price the container reference build. For VMs, include hypervisor support, guest licences and VM backup capacity, and replace the reference operating hours with your measured budget in Move 03.

## COMPUTE

- ☐ 16 1U dual-socket servers, and 2 on the shelf
- ☐ 32 physical cores a node, current-generation EPYC or Xeon
- ☐ 256 GB ECC memory a node, all channels populated
- ☐ Redundant hot-swap power supplies, each fed from a different strip
- ☐ A management controller on every node - Redfish, not just a web console

## STORAGE

- ☐ Four NVMe drives a node, mixed-use endurance, 3.84 TB each
- ☐ Enterprise drives with power-loss protection - not consumer SSDs
- ☐ A separate boot pair, mirrored
- ☐ One spare drive of each size, on site, unopened
- ☐ A backup target at a provider you are not otherwise using

## NETWORK AND SPACE

- ☐ Two 25 GbE switches, stacked or in a link-aggregation pair
- ☐ Two uplinks a node, one to each switch
- ☐ A separate 1 GbE management switch, not reachable from the internet
- ☐ A full rack in a carrier-neutral facility, A and B power
- ☐ Facility transit for go-live, and a second upstream ordered alongside it
- ☐ An out-of-band line that works when the main path does not

## THE SOFTWARE STACK

- ☐ Proxmox VE on VM hosts, with KVM guests for Linux and Windows workloads
- ☐ Talos Linux for Kubernetes nodes, on hardware or in KVM guests
- ☐ Kubernetes, on a version at least one minor behind the newest
- ☐ Cilium for the cluster network, with kube-proxy replaced
- ☐ Ceph owned by Proxmox for VM disks or by Rook on direct Talos nodes
- ☐ CloudNativePG, on local NVMe rather than on Ceph
- ☐ cert-manager and Envoy Gateway
- ☐ Argo CD, with the cluster state in Git
- ☐ Prometheus and Loki for the stores, OneUptime for what wakes somebody
- ☐ Velero for Kubernetes; separate whole-VM backups, each restore rehearsed

## ACCESS

- ☐ An identity provider that is not the cluster
- ☐ Break-glass credentials in a safe, on paper
- ☐ A jump host or a mesh - never a public API server
- ☐ Hardware tokens for anyone who can delete a cluster
- ☐ A status page that is not hosted on the thing it reports on
- ☐ The domain registrar account locked, with recovery codes printed

## ON THE LAPTOP

- ☐ kubectl
- ☐ helm
- ☐ talosctl
- ☐ k9s
- ☐ argocd
- ☐ velero
- ☐ psql and pg\_dump
- ☐ barman-cloud
- ☐ redis-cli
- ☐ rclone
- ☐ the cloud CLI, still logged in
- ☐ dig and mtr
- ☐ openssl
- ☐ ipmitool
- ☐ the runbook, printed



# THE ON-RAMP

Almost nobody should sign a facility contract before they have run this stack once. A cluster of 3 refurbished machines with 12 cores and 64 GB each, on a managed switch under a desk, lets you rehearse the chosen platform and a service migration. Adapt the disk layout to the smaller kit. It costs about \$3,430 once and about \$30 a month in electricity, it saves nothing whatsoever, and it is the cheapest way to find out whether the rest of this book is for you.

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## WHY THREE, AND WHY SECOND-HAND

Three because a control plane needs three members to lose one and keep going, and everything you will learn about quorum, fencing and upgrades needs somewhere to go wrong. Second-hand because a machine two generations old runs Kubernetes exactly as well as a new one at roughly a tenth of the price, and because you want to be willing to break it.

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## WHAT IT CANNOT TEACH YOU

Every word of Stage 2. A homelab has one power feed, one switch, no cross-connect, no remote hands and nobody to escalate to at three in the morning. It cannot show you what a colocation contract is for, or what happens when the A feed goes away and the B feed was never tested. The building is a different discipline, which is why it has a stage of its own.

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## THE ONE PART TO BUY NEW

The drives. A consumer SSD without power-loss protection will corrupt etcd the first time the power flickers, and it will do it in a way that looks like a Kubernetes bug for a day and a half. Buy small enterprise NVMe with the protection, second-hand if you like, and put the money you saved on the chassis into the disks.

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## WHAT IT PROVES

For VMs, Proxmox guest provisioning, networking and a backup restored on another host. For containers, Talos, Kubernetes and a Postgres cutover against a copy. Rehearse the storage branch you intend to run; this small kit does not prove production capacity or recovery time.

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## WHEN TO STOP USING IT

Never. The on-ramp becomes the non-production cluster the rest of the book keeps asking for — the one you restore etcd onto, rehearse an upgrade on, and point a load generator at. Every estate needs a cluster it is allowed to destroy, and this is the cheapest one you will ever own.

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## IF THREE MACHINES AT HOME IS NOT POSSIBLE

Rent three suitable dedicated hosts available on monthly terms and run the same experiment. It costs more than electricity and less than being wrong, it gives you real addresses and a real network, and you can end the rental under the provider's notice terms. What it will not give you is a machine you can physically pull a disk out of.

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## THE WHOLE SHOPPING LIST

- ☐ Three refurbished mini PCs or small-form-factor desktops
- ☐ 64 GB in each - memory is what runs out first, not cores
- ☐ Two NVMe drives a node: one to boot, one for data
- ☐ A managed switch with VLANs and a spare port for out-of-band
- ☐ A small uninterruptible power supply, mostly to survive the brownouts that corrupt etcd
- ☐ A domain you own, for real certificates rather than self-signed ones

# THE RULES FOR LEAVING THE CLOUD

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- 1 **Move the cheapest thing first.** Not the most expensive. The first Move exists to prove the platform and the team, and it should be something whose failure is a shrug. You are buying evidence, and evidence is worth more early than savings are.

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  - 2 **Data moves last and leaves last.** Every stateless thing can be cut over and cut back in minutes. State cannot. Move compute first, run it against the cloud database across the link, and only then move the database - by which point you have already learned the things you would otherwise learn during a data migration.

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  - 3 **Never run two platforms longer than you planned to.** The cost of leaving is not the hardware. It is the months in which both estates are live, both are on call, and every change has to be made twice. Pick the overlap window before you start, write it down, and treat overrunning it as the incident it is.

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  - 4 **Keep the bill until the invoice proves you can stop.** A resource is not decommissioned when nothing points at it. It is decommissioned when a full month has passed, the line has gone from the bill, and nobody has filed a ticket. Turning things off too early is how a migration becomes an outage a fortnight later.

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  - 5 **Buy the second of everything, and one spare.** Two switches, two power feeds, two people who can get into the building, and spare machines that are already burned in. Every single point of failure you accept at the start becomes an incident you attend in person, at night, in a year.

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  - 6 **Count the salary.** A saving that costs engineer-hours is a real saving only if you write the hours into the comparison - on both sides, because the cloud takes them too. Most of these still win by a wide margin once you do. The ones that do not, you want to know about in week one rather than month ten.

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  - 7 **Keep paying for the three things you will not beat.** A content delivery network, denial-of-service scrubbing at the edge, and outbound email deliverability. Each is a business somebody else already runs better than you will, and each is cheap next to what it replaces. Ownership is a means, not a principle.

# BEFORE YOU TOUCH ANYTHING

Every Move in this book changes something that is running. Most are reversible for a stated window, a few only until a particular step, and none of them is irreversible, which is a property of this edition rather than of the work: every Move here was chosen so that it could be undone. The difference between a migration and an incident is almost never the technique - it is whether somebody worked out the way back before they started, and whether anybody had ever tested it.

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## READ THE ROLLBACK FIRST

Not the runbook. The Rollback section of every Move names the point of no return: the step after which the old system can no longer serve traffic or no longer holds current data. Read it, decide whether you are willing to reach that step today, and only then start at step one.

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## KEEP THE OLD THING RUNNING, AND KEEP PAYING FOR IT

The instinct after a successful cutover is to delete the source and book the saving. Do not. The Reversible field on each Move is how long the old system must stay warm, powered and receiving its own backups. That overlap is the entire cost of being wrong, and it is cheap.

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## WRITE DOWN THE STATE YOU CANNOT RECREATE

Before Stage 4, list every store whose contents exist nowhere else: the primary database, the uploads bucket, the secrets, the certificate private keys, the recovery codes for the accounts that own the domain. Everything else is rebuildable from a repository. That list is short, and it is the only part of the estate where a mistake is permanent.

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## TWO PEOPLE, ONE KEYBOARD

Every High-risk Move in this book assumes a second person who is not typing, is reading the runbook aloud, and has the authority to stop. It is the cheapest safety control in infrastructure and the first one teams drop when they are behind schedule. With two engineers on the whole programme, it is also the one you are most tempted to skip.

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## A BACKUP NOBODY RESTORED IS NOT A BACKUP

It is a file. Before Stage 4, restore the backup you are relying on into a scratch environment and count the rows against production. The number of companies that discover their backups were empty during the incident that needed them is not small, and every one of them had a green dashboard.

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## CUT OVER WHEN YOU CAN UNDO IT, NOT WHEN YOU ARE CONFIDENT

Confidence is not evidence. The right time to move traffic is when the rollback has been rehearsed end to end, on the same day of the week, with the same people awake. A Tuesday morning with the team at their desks beats a Saturday night with one person and a laptop, every time.

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## VERIFY BY COUNTING, NOT BY LOOKING

A migration is finished when the destination answers the same questions as the source with the same answers. Row counts, checksums, a replayed hour of real traffic compared response by response. A page that loads is not evidence, and neither is an application that starts.

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## STOP WHEN YOU ARE SURPRISED

Not when something breaks - when something is merely unexpected. A row count that is off by eleven, a certificate with the wrong issuer, a replica that caught up faster than it should have. Surprise means the model in your head and the system in front of you have diverged, and continuing is guessing.

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These are operational practices, not a warranty. The runbooks in this book were written against the services and versions named in each Move and will drift as those change. Rehearse every Move against a copy of your own estate before you run it against the real one, and treat any figure here as the shape of an answer rather than a quotation for your account.

# WHAT IT ACTUALLY COSTS

The existing operations team carries 160 hours a month in every option; physical support stays in the provider charges. Substitute your rates and hours. These figures price the container reference build. For VMs, include hypervisor support, guest licences and VM backup capacity, and replace the reference operating hours with your measured budget in Move 03.

## ON AWS, COMPUTE ALONE

1,024 vCPU of current-generation general purpose instances, on demand, is **\$37,675** a month before a single gigabyte of storage, a load balancer or a byte of egress. Egress is the line that ends most arguments: 100 TB a month is \$7,980, every month, for the privilege of your own traffic leaving.

## RENTED BY THE MONTH

\$11,700 buys the reference fleet from a dedicated-host provider, with the machines, power, space, network and physical maintenance in the rental. You keep the capital available and avoid a separate facility lease. With retained services and no increase in operations hours, the cloud bill line becomes **\$20,880** a month, base salary unchanged. Available stock can shorten the move. Confirm delivery, bandwidth, support and notice terms in the quote. Rent when preserving cash matters more than owning the fleet.

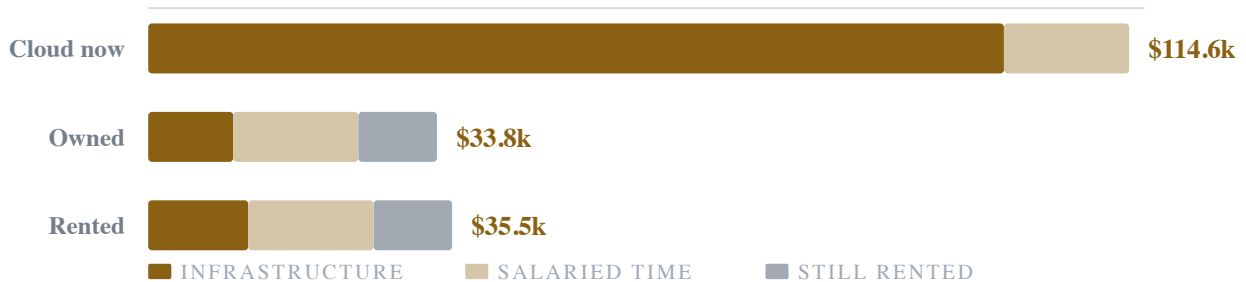
## OWNED, IN COLOCATION

\$9,935 of infrastructure covers metal amortised over five years, the spares, power, a full rack, transit, the cross-connect and remote hands. Add \$0 of additional salaried time and \$9,180 of retained services: the edge, outbound mail, scrubbing and the residue of later Moves. Those three are **\$19,115** a month: what the \$100,000 line becomes, base salary unchanged on both sides. Remote hands performs the agreed physical interventions; your existing team runs the platform and applications within the same monthly hours.

## WHAT THE DIFFERENCE BUYS

\$80,885 a month, or \$970,620 a year, for a capital outlay of \$248,000 — 18 machines and a pair of switches — which pays for itself in about 3 months. That is 71 per cent of a fully loaded \$114,644, with the salary counted on both sides; on the infrastructure line alone, which is what every other comparison quotes, it is 81 per cent. The dollar saving is the same; counting the existing salary increases the baseline.

## ONE ESTATE, THREE WAYS, PER MONTH



Every AWS figure is a public list price for us-east-1 observed while writing, before any Savings Plan, private pricing agreement or credit. Every hardware figure is a mid-market street price for the specification opposite. Both will drift. The comparison is a method you can repeat against your own invoice, not a quotation - and if your effective rate is half of list, halve the left-hand column and read the conclusion again.

# FIVE YEARS, THREE WAYS

Both metal options cut the cost of this reference estate over the life of the machines. Renting keeps the server purchase off the opening cash flow; colocation trades that purchase for lower monthly infrastructure costs. The table separates capital from running costs and counts the same operations hours in all three options, so you can choose the cash flow that suits the business.

THE SAME COMPUTERS, THREE WAYS, OVER 5 YEARS

|              | CAPITAL   | RUNNING, A MONTH | TOTAL OVER 5 YEARS | AGAINST THE CLOUD |
|--------------|-----------|------------------|--------------------|-------------------|
| Cloud        | —         | \$114,644        | \$6,878,613        | —                 |
| Rented metal | —         | \$35,524         | \$2,131,413        | \$4,747,200 (69%) |
| Colocation   | \$248,000 | \$29,692         | \$1,992,293        | \$4,886,320 (71%) |

In this model, colocation costs \$139,120 less than rented metal over five years. That includes lower infrastructure costs and \$37,200 of estimated fleet value at the end, counted at fifteen per cent of the purchase price. Renting preserves that opening capital and puts physical maintenance with the provider. Replace both quotes and the residual estimate with your own; location, bandwidth and support can change which route wins.

# WHAT REPLACES WHAT

Every Move names the real service on all three clouds and the one thing that differs on each, so this table is the map rather than the content. Find the row you are paying for and then read the Move. The right-hand column is the point of the whole book, and where it says keep paying, that is a conclusion rather than a gap.

| AWS                              | GOOGLE CLOUD             | AZURE                      | WHAT YOU RUN INSTEAD                                                    |
|----------------------------------|--------------------------|----------------------------|-------------------------------------------------------------------------|
| <b>EC2</b>                       | Compute Engine           | Virtual Machines           | KVM virtual machines on Proxmox VE hosts                                |
| <b>EC2 Auto Scaling</b>          | Managed instance groups  | Virtual Machine Scale Sets | Fixed capacity, a priority class, and a spare on the shelf              |
| <b>EKS</b>                       | GKE                      | AKS                        | Talos and Kubernetes, on hardware or in Proxmox VE guests               |
| <b>ECS on Fargate</b>            | Cloud Run                | Container Apps             | Deployments and Jobs on your own nodes                                  |
| <b>AMI</b>                       | Machine images           | Managed images             | KVM guest templates; rebuild or rehearse a supported disk import        |
| <b>VPC</b>                       | VPC networks             | Virtual Network            | Two switches, four ranges, and an address plan on one page              |
| <b>EBS</b>                       | Persistent Disk          | Managed Disks              | Proxmox VM disks on Ceph RBD; Rook volumes for Kubernetes               |
| <b>EFS</b>                       | Filestore                | Azure Files                | CephFS under Rook, or a filesystem the application stops needing        |
| <b>S3</b>                        | Cloud Storage            | Blob Storage               | Ceph's S3 gateway - deep archive stays where it is                      |
| <b>RDS for PostgreSQL</b>        | Cloud SQL for PostgreSQL | Database for PostgreSQL    | PostgreSQL under CloudNativePG, on local NVMe                           |
| <b>ElastiCache</b>               | Memorystore              | Azure Cache for Redis      | Valkey for the half that is really a cache                              |
| <b>SQS and SNS</b>               | Pub/Sub                  | Service Bus                | NATS JetStream, or a table in the database you already have             |
| <b>Glacier Deep Archive</b>      | Archive storage class    | Archive tier               | <b>Keep paying - the egress to extract it costs more than the years</b> |
| <b>AWS Backup</b>                | Backup and DR Service    | Azure Backup               | Whole-VM backups, Velero and database archives, held off-site           |
| <b>Application Load Balancer</b> | Cloud Load Balancing     | Application Gateway        | Envoy Gateway behind a virtual IP that fails over                       |

# WHAT REPLACES WHAT II

The rows that say keep paying are set in red because they are the most important lines in the table. A content delivery network, scrubbing capacity at the edge and outbound mail deliverability are not technical problems you have not solved yet. They are businesses, and you are not in them.

| AWS                       | GOOGLE CLOUD              | AZURE                         | WHAT YOU RUN INSTEAD                                               |
|---------------------------|---------------------------|-------------------------------|--------------------------------------------------------------------|
| <b>ACM</b>                | Certificate Manager       | Key Vault certificates        | cert-manager, issuing and renewing without anybody watching        |
| <b>Route 53</b>           | Cloud DNS                 | Azure DNS                     | Two independent authoritative providers, never one                 |
| <b>NAT Gateway</b>        | Cloud NAT                 | NAT Gateway                   | Node addresses, and one translation hop for the few that need it   |
| <b>CloudFront</b>         | Cloud CDN                 | Azure Front Door              | <b>Keep paying - change vendor, not model</b>                      |
| <b>AWS Shield and WAF</b> | Cloud Armor               | Azure DDoS Protection and WAF | <b>Keep paying for scrubbing - the capacity is upstream of you</b> |
| <b>SES</b>                | No first-party equivalent | Communication Services        | <b>Keep paying for outbound - inbound can come home</b>            |
| <b>ECR</b>                | Artifact Registry         | Container Registry            | A pull-through cache, then your own registry of record             |
| <b>Secrets Manager</b>    | Secret Manager            | Key Vault                     | Encrypted in the repository, with the key held outside the cluster |
| <b>CodeBuild</b>          | Cloud Build               | Azure Pipelines               | Runners on the machines you already own and are not using          |
| <b>CloudWatch metrics</b> | Cloud Monitoring          | Azure Monitor                 | Prometheus, with a cardinality budget agreed in advance            |
| <b>CloudWatch Logs</b>    | Cloud Logging             | Log Analytics                 | Loki, sized against retention somebody actually chose              |
| <b>CloudWatch alarms</b>  | Cloud Monitoring alerting | Azure Monitor alerts          | OneUptime - alerting, on-call rotas, incidents and a status page   |
| <b>Cost Explorer</b>      | Cloud Billing reports     | Cost Management               | A model built from invoices, counting power, spares and the rota   |
| <b>Organizations</b>      | Organisation and projects | Subscriptions and Entra       | One account left open only until the last invoice reads zero       |

# WORK OUT WHETHER TO DO IT AT ALL

DONE WHEN You know what you spend, what you would spend instead, and whether it is worth it.

Whether to do this at all. Three Moves of arithmetic against your own invoice, and a fourth that names what stays rented. The stage most likely to end the programme, and that is a success rather than a failure.

---

|                          |    |                                                   |       |    |
|--------------------------|----|---------------------------------------------------|-------|----|
| <input type="checkbox"/> | 01 | The bill, and the three lines that are most of it | 0 min | 24 |
|--------------------------|----|---------------------------------------------------|-------|----|

---

|                          |    |                       |       |    |
|--------------------------|----|-----------------------|-------|----|
| <input type="checkbox"/> | 02 | What you actually run | 0 min | 26 |
|--------------------------|----|-----------------------|-------|----|

---

---

|                          |    |                            |       |    |
|--------------------------|----|----------------------------|-------|----|
| <input type="checkbox"/> | 03 | The number that decides it | 0 min | 28 |
|--------------------------|----|----------------------------|-------|----|

---

|                          |    |                                   |       |    |
|--------------------------|----|-----------------------------------|-------|----|
| <input type="checkbox"/> | 04 | The three things you keep renting | 0 min | 30 |
|--------------------------|----|-----------------------------------|-------|----|

---



# ORDER THE HARDWARE AND SIGN THE SPACE

DONE WHEN The machines are on order and the cage is signed.

What to order and where to put it. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy. Both clocks start in this stage, and they run in parallel.

---

|                          |    |                                            |       |    |
|--------------------------|----|--------------------------------------------|-------|----|
| <input type="checkbox"/> | 05 | From rented vCPUs to cores you own         | 0 min | 34 |
| <hr/>                    |    |                                            |       |    |
| <input type="checkbox"/> | 06 | Sixteen machines, and the two on the shelf | 0 min | 36 |

---

---

|                          |    |                                              |       |    |
|--------------------------|----|----------------------------------------------|-------|----|
| <input type="checkbox"/> | 07 | A cage, not a data centre                    | 0 min | 38 |
| <hr/>                    |    |                                              |       |    |
| <input type="checkbox"/> | 08 | The order, and the weeks you cannot compress | 0 min | 40 |

---

# TURN THE BOXES INTO A CLUSTER

DONE WHEN A cluster that could take production traffic, and never has.

Everything between a racked machine and a cluster that could take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

---

|                          |                    |                                                 |       |    |                          |                    |                                           |       |    |
|--------------------------|--------------------|-------------------------------------------------|-------|----|--------------------------|--------------------|-------------------------------------------|-------|----|
| <input type="checkbox"/> | <a href="#">09</a> | Racking day                                     | 0 min | 44 | <input type="checkbox"/> | <a href="#">11</a> | The platform your workloads need          | 0 min | 48 |
| <input type="checkbox"/> | <a href="#">10</a> | The network, and the way back in when it breaks | 0 min | 46 | <input type="checkbox"/> | <a href="#">12</a> | Disks: what goes local, what goes on Ceph | 0 min | 50 |

---

# MOVE THE APP, THEN THE DATA

DONE WHEN Everything runs on your machines, and the cloud copy is still warm.

The application first, then the state. The stage that can lose a company its data, and therefore the one with the longest overlap windows, the most rehearsal and the fewest one-way steps.

|                          |    |                                         |        |    |
|--------------------------|----|-----------------------------------------|--------|----|
| <input type="checkbox"/> | 13 | Images, secrets and one-command deploys | 0 min  | 54 |
| <input type="checkbox"/> | 14 | The first service, end to end           | 0 min  | 56 |
| <input type="checkbox"/> | 15 | Buckets, cache and queues               | 0 min  | 58 |
| <input type="checkbox"/> | 16 | Postgres, the one that matters          | 15 min | 60 |

# CUT THE TRAFFIC OVER, AND KEEP IT ALIVE

**DONE WHEN** Users reach your machines, you can carry it at 03:00, and the cloud bill is zero.

The traffic, the pager and the last account. Every article about leaving the cloud stops before this stage, and every migration that regrets itself failed inside it.

|                          |    |                                          |        |    |
|--------------------------|----|------------------------------------------|--------|----|
| <input type="checkbox"/> | 17 | The front door                           | 0 min  | 64 |
| <input type="checkbox"/> | 18 | Go-live, and how you abort               | 10 min | 66 |
| <input type="checkbox"/> | 19 | Backups you have restored, and the pager | 0 min  | 68 |
| <input type="checkbox"/> | 20 | Closing the account                      | 0 min  | 70 |

## DECIDE

The invoice, the inventory and the comparison with the salary line in it. Three Moves, about a week, and permission to stop.

## BUY

The specification, the machines, the space and everything with a lead time. The stage where a mistake costs a lorry rather than a deploy.

## BUILD

Racking, the network, the cluster and the disks. Nothing here moves a workload, and everything here decides how the rest goes.

## MOVE

Images and secrets, the first service, the buckets and the database. The longest overlaps and the most rehearsal in the book.

## RUN

The edge, the cutover, the backups you have restored, the pager, and the account you finally close.

Every Move states its cutover in minutes of user-visible downtime, its risk as blast radius rather than difficulty, and how long the thing it replaced must stay warm before you turn it off. 18 of the 20 Moves in this book cost no downtime at all; the whole book, executed end to end, costs 25 minutes.

# WHICH MOVE DO I NEED

Nobody opens a book like this at Move 01. They open it because something on the bill, or on the pager, has become intolerable. This is the index for the way the question actually arrives.

---

## The bill went up and nobody can say why

- 01 The bill, and the three lines that are most of it  
 02 What you actually run    03 The number that decides it
- 

## Egress is now one of the largest lines on the invoice

- 01 The bill, and the three lines that are most of it  
 14 The first service, end to end    18 Go-live, and how you abort
- 

## We have no idea what we are actually running

- 02 What you actually run  
 13 Images, secrets and one-command deploys    20 Closing the account
- 

## The hardware sounds like a full-time job nobody has

- 07 A cage, not a data centre    09 Racking day  
 11 The platform your workloads need
- 

## We want remote hands to handle the hardware calls

- 07 A cage, not a data centre  
 19 Backups you have restored, and the pager  
 06 Sixteen machines, and the two on the shelf
- 

## Somebody has asked what would happen if we just left

- 03 The number that decides it    04 The three things you keep renting  
 07 A cage, not a data centre
- 

## The database costs more than the engineers who query it

- 05 From rented vCPUs to cores you own  
 12 Disks: what goes local, what goes on Ceph  
 16 Postgres, the one that matters
- 

## We are being asked to do this and we have two engineers

- 03 The number that decides it  
 06 Sixteen machines, and the two on the shelf  
 19 Backups you have restored, and the pager
- 

## Something has to move this quarter and nothing may break

- 14 The first service, end to end    17 The front door  
 18 Go-live, and how you abort
- 

## We were told we cannot leave, and nobody has checked

- 04 The three things you keep renting    15 Buckets, cache and queues  
 20 Closing the account
-



# I DECIDE

Whether to do this at all. Three Moves of arithmetic against your own invoice, and a fourth that names what stays rented. The stage most likely to end the programme, and that is a success rather than a failure.

|            |                    |                |                     |
|------------|--------------------|----------------|---------------------|
| 4<br>MOVES | 4<br>ZERO DOWNTIME | 0<br>HIGH RISK | 0<br>MINUTES, TOTAL |
|------------|--------------------|----------------|---------------------|

IF YOU ONLY DO THREE

01

## THE BILL, AND THE THREE LINES THAT ARE MOST OF IT

The invoice records what you were charged, not what you use. Read it to the line, find the three that are most of it ...

04

## THE THREE THINGS YOU KEEP RENTING

Three capabilities do not come home: the edge, outbound mail and volumetric scrubbing. Naming them now keeps the plan ...

02

## WHAT YOU ACTUALLY RUN

The estate feels infinite because nobody has written it down. On one page it is forty rows, and every row has a name ...

THE BILL, AND THE THREE LINES THAT ARE MOST OF IT · WHAT YOU ACTUALLY RUN · THE NUMBER THAT DECIDES IT · THE THREE THINGS YOU KEEP RENTING

■ 3 LOW RISK

■ 1 MEDIUM RISK

# 01 THE BILL, AND THE THREE LINES THAT ARE MOST OF IT

CUTOVER

0 MIN



|                 |              |             |                           |
|-----------------|--------------|-------------|---------------------------|
| LAYER<br>DECIDE | LEAVING<br>— | RISK<br>LOW | REVERSIBLE<br>IMMEDIATELY |
|-----------------|--------------|-------------|---------------------------|

The invoice records what you were charged, not what you use. Read it to the line, find the three that are most of it, then start the clock on working set.

|        |                              |                                                                                                                                              |
|--------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | <b>Cost Explorer</b>         | hourly and resource-level granularity is a paid opt-in, and that detail is then kept for 14 days.                                            |
| GOOGLE | <b>Cloud Billing reports</b> | detailed usage cost reaches BigQuery only from the day the export is enabled, and is never backfilled.                                       |
| AZURE  | <b>Cost Management</b>       | amortised reservation cost appears on an Enterprise Agreement account, not on pay-as-you-go, and exports land in Blob Storage on a schedule. |

## BEFORE YOU START

### ACCESS

- Billing owner rights in every cloud account, and permission to install an agent on production instances

### SOFTWARE

- The provider tools aws, gcCloud and az, none older than six months
- A spreadsheet and a time-series store with three weeks of retention

### PEOPLE

- Whoever signs the invoice, for half an hour at the end

## WHY THIS WORKS

Three numbers decide everything that follows, and none is on the provider's default dashboard. A bill records what was charged — allocation, commitment, a support percentage — not what the machines were doing. Grouped into compute, state and traffic, two hundred line items collapse into three piles, and three lines usually carry eighty per cent of the spend. The window runs alongside, because the figure a purchase order needs is working set and the hypervisor cannot see it. Two weeks, not thirty days: this is a five-machine decision, and more precision would not change the order.

### SWAP

If finance already breaks the bill down by service, start there and spend the saved day on step four. The categories will be wrong, the totals right.

### DO IT FASTER

Pull three invoices rather than one. One month hides a reservation renewal and a support tier that stepped up.

### WATCH OUT

Agent metrics are billed differently on each cloud — per unique metric a month on one, per sample ingested on the others — and an agent collecting every counter it finds has produced four-figure surprises on all three. Name the metrics you want.

### LEFTOVERS

The agents, export jobs and extra ingestion bill on until somebody removes them. Diary it for the day the hardware order is signed.

|          |          |            |                  |                  |                 |
|----------|----------|------------|------------------|------------------|-----------------|
| —<br>WAS | —<br>NOW | —<br>SAVED | 0 min<br>CUTOVER | 3 days<br>EFFORT | 2 weeks<br>WAIT |
|----------|----------|------------|------------------|------------------|-----------------|

TURN OFF Nothing, yet. This Move moves one line upward: a fortnight of custom metrics, the cheapest few hundred dollars in the programme.



- 01 Export one month at line-item granularity from every account, not the summary view. That means the detailed export rather than the console API on all three: the Cost and Usage Report delivered to a bucket, a BigQuery billing export, and `az costmanagement export` per subscription. `aws ce get-cost-and-usage` and the reports page return period totals grouped two ways at most, which is the summary view under another name.
- 02 Put every line in the spreadsheet, one row each, tagged compute, state or traffic: instances and containers; managed databases, caches and object storage; egress, balancers and the CDN.
- 03 Sort by amount and rule a line under the third row. Those three are the programme; anything beneath worth under two per cent of the invoice is not discussed again.
- 04 Reconcile the export against what you believe you run. Enumerate the accounts first — `aws organizations list-accounts`, `gcloud billing projects list` against each billing account, `az account list` — because the forgotten estate is usually in an account nobody named. Lines nobody owns are idle balancers, volumes of dead instances, a staging project that shipped. Expect five to fifteen per cent.
- 05 Install the memory agent where it is missing and point one time-series store at all three metric APIs. No hypervisor sees inside its guests, so working set arrives only from within the machine.
- 06 Open the window for a fortnight, and record what resists measurement: burstable credits, page cache against anonymous memory, build minutes that were never an instance.

## ROLLBACK

Nothing in production changes, so backing out is removing an agent and stopping an export job. There is no point of no return here; the first is the hardware order, signed against these numbers. A sampling window cannot be rerun retrospectively, so stopping at day nine means starting over. If the time-series store is lost, restore it.



# 02 WHAT YOU ACTUALLY RUN

CUTOVER

0 MIN



|                 |              |             |                           |
|-----------------|--------------|-------------|---------------------------|
| LAYER<br>DECIDE | LEAVING<br>— | RISK<br>LOW | REVERSIBLE<br>IMMEDIATELY |
|-----------------|--------------|-------------|---------------------------|

The estate feels infinite because nobody has written it down. On one page it is forty rows, and every row has a name against it.

|        |                       |                                                                                                                                                                          |
|--------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | Resource Explorer     | indexes per Region behind one aggregator and searches one account until the organisation-wide view is on, so the second account stays invisible.                         |
| GOOGLE | Cloud Asset Inventory | the caller needs <code>cloudasset.assets.searchAllResources</code> at the organisation scope; a project-level grant says nothing about the siblings on the same invoice. |
| AZURE  | Resource Graph        | reads only subscriptions the caller can already see, and pages at 1,000 rows, so an unpaginated query returns a truncated estate that looks complete.                    |

## BEFORE YOU START

### ACCESS

- Read-only access to every cloud account, project and subscription, including the forgotten ones
- The billing export from Move 01, at resource granularity rather than by service

### SOFTWARE

- The provider's asset CLI — `aws`, `gcloud` or `az` — and one shared document with a row per resource

### PEOPLE

- The engineer who would notice each service stopping, named on every row

## WHY THIS WORKS

Leaving the cloud feels impossible because nobody can say what would have to move. The estate lives in three or four heads, so the total is always imagined larger than it is. A company spending \$100,000 a month usually runs forty or fifty services, a handful of Postgres clusters, two or three caches, a dozen buckets and a scheduler nobody has read in a year. That still fits on a page, and every later Move then has something finite to count against. Nothing changes here: this Move reads, and writes down what it read. The risk is being wrong about what runs, and the bill settles that.

### SWAP

Where accounts are few and tags honest, the provider's tag report is a quicker first pass. Untagged resources still have to be found by hand.

### DO IT FASTER

Reconcile before assigning owners. The bill sorts rows by cost, which is the order worth arguing about.

### WATCH OUT

These inventories list what the provider created. What a third party stood up against your account — an agent, a contractor's sandbox — appears nowhere else.

### LEFTOVERS

A snapshot outlives the volume it came from and bills long after the rest of its account has gone. So does a retained log group.

|     |     |       |         |        |      |
|-----|-----|-------|---------|--------|------|
| —   | —   | —     | 0 min   | 4 days | —    |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** The fourth pile, once its owners confirm: orphaned volumes, snapshots of vanished disks, idle load balancers and the proof-of-concept accounts. The cheapest saving in the book.

**NEEDS** 01 The bill, and the three lines that are most of it

- 01 Pull the machine-readable list, one account at a time: `aws resource-explorer-2 search`, `gcloud asset search-all-resources` or `az graph query`. Paste it into one document, a row per resource.
- 02 Add what those APIs never return: scheduled jobs, queues, third-party services with a webhook into production, and the DNS records pointing at any of it.
- 03 Reconcile against the billing export from Move 01 both ways. A bill line with no row is unowned; a row with no bill line is free, or in an account nobody has opened.
- 04 Hunt what hides: snapshots whose disk is gone, volumes attached to nothing, load balancers with no target, staging at full size overnight, the accounts opened for a proof of concept in 2022.
- 05 Label each workload VM, container or managed service. For each VM record its operating system, CPU architecture, boot mode, attached disks, licences and metadata credentials; mark which disks contain persistent state. A VM stays a VM on Proxmox VE in Move 11. Moving an application into containers is a separate project, not an unstated prerequisite.
- 06 Sort every row into one of four piles — moves as-is, moves with work, stays rented, gets deleted. The fourth is the biggest; confirm each row with its owner and check thirty days of metrics first. Record a destination and a data-transfer method for every row that moves; image export is only an option after its restrictions are checked.

## ROLLBACK

Nothing here touches a running system, so backing out is discarding a document nobody acted on. The point of no return is the fourth pile: once a resource on it is deleted, getting it back means a restore from whatever snapshot survives, and for a log group there may be none. Keep the inventory in version control.



# 03 THE NUMBER THAT DECIDES IT

CUTOVER

0 MIN



|        |         |        |             |
|--------|---------|--------|-------------|
| LAYER  | LEAVING | RISK   | REVERSIBLE  |
| DECIDE | —       | MEDIUM | IMMEDIATELY |

Three columns, the same operations team and five years of costs. Put colocation and rented metal beside the cloud bill and choose the margin worth moving for.

|        |                                        |                                                                                                                                                        |
|--------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | <b>AWS Pricing Calculator</b>          | two tools now carry that name, and only the one inside the billing console can see the Savings Plans your account already holds.                       |
| GOOGLE | <b>Google Cloud Pricing Calculator</b> | the only discount it models is a committed-use one you pick from a dropdown, and its default machine series earns no sustained-use discount at all.    |
| AZURE  | <b>Azure Pricing Calculator</b>        | signing in and selecting the Enterprise Agreement licensing programme rewrites every price on the page, so an anonymous quote is retail and not yours. |

## BEFORE YOU START

### ACCESS

- Three recent invoices at the rate you actually paid, not list
- Quotes for sixteen active machines, two spares and a full rack, plus an equivalent dedicated-server rental

### SOFTWARE

- One spreadsheet, three columns: cloud, rented metal and colocation

### PEOPLE

- The existing cloud operations lead, to map the team's work onto the new platform

## WHY THIS WORKS

Moves 01 and 02 produced a grouped bill and an inventory that accounts for it. This Move compares three ways to run that estate: the cloud, rented dedicated servers and owned hardware in colocation. Both metal options can keep the existing operations team while reducing the infrastructure bill; renting also keeps the hardware purchase off the opening budget. Put support, retained services and one-time migration work into the comparison, then apply a margin chosen before the figures. Two days of analysis gives the team a decision it can fund and defend.

### SWAP

Price the three largest lines only; that lands within about ten per cent in an afternoon.

### DO IT FASTER

Use one sheet with the same capacity, support and retained-service assumptions in every column; compare it with the cloud rate you actually pay.

### WATCH OUT

Equal recurring hours assumes an existing production team, a standardised platform and contracted physical support. For VMs, include hypervisor updates, guest OS patching and restore drills when validating that assumption. Vary hours, rental rates and hardware life in the sheet; setup and migration are one-time work.

### LEFTOVERS

The committed-spend agreement bills to the end of its term, and that renewal date is the deadline.

|     |     |       |         |        |      |
|-----|-----|-------|---------|--------|------|
| —   | —   | —     | 0 min   | 2 days | —    |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT |

TURN OFF Nothing yet. This Move produces a number and, with it, permission to act or to stop.

NEEDS 01 The bill, and the three lines that are most of it 02 What you actually run

## THE RUNBOOK

- 01 Write the decision rule in the spreadsheet before any figure goes in: a metal option must land at least a third under the current cloud cost to justify this programme. Apply the same margin to rented metal and colocation.
- 02 Fill the cloud column from the three invoices at the rate you actually paid, discounts in and tax out across all columns. One monthly number, checkable against Move 01.
- 03 Fill the colocation column from hardware and facility quotes: sixteen machines and two spares over sixty months, rack, power, transit, cross-connects and remote hands. Quote equivalent rented metal with traffic and hardware support. For VMs, add the chosen Proxmox support subscription, guest OS and application licences, and off-site VM backup capacity to each applicable column. The published figures describe the container reference estate; price your VM mix separately.
- 04 Carry the existing operations salary into all three columns. This reference model budgets the same 160 hours a month for each: cloud ops transitions to platform ops, with physical work assigned to the facility's remote hands or the rental provider under the support contract. Validate that assumption with the pilot and the people doing the work. Price setup, training and migration once in the project budget; any measured recurring difference belongs in the monthly line.
- 05 Add the services that stay rented. Move 04 identifies the edge, outbound mail and scrubbing; later Moves retain archived object storage, a registry, a queue and an off-site backup copy. Their cost is already in the cloud invoice and must remain in both metal columns.
- 06 Total the three columns and apply the rule. If either metal option clears it, compare the five-year totals and the monthly cash requirement. If neither clears, keep the sheet and revisit it when the estate or quotes change.
- 07 Choose how to fund the move. Colocation buys the fleet and carries its value forward; rented metal pays monthly and keeps that capital available. Put the actual purchase quote, one-time migration effort and overlap with the cloud bill in front of whoever signs. Move 07 settles the facility and rental terms.

## ROLLBACK

Nothing is ordered and nothing signed, so this Move reverses by changing a cell. The point of no return is the order in Move 06 and the contract in Move 07. Keep dated versions of the sheet, so earlier assumptions can be restored when somebody asks how the decision was reached.



# 04 THE THREE THINGS YOU KEEP RENTING

CUTOVER

0 MIN



|                 |                                                          |             |                           |
|-----------------|----------------------------------------------------------|-------------|---------------------------|
| LAYER<br>DECIDE | LEAVING<br>NOTHING — THIS MOVE DECIDES WHAT STAYS RENTED | RISK<br>LOW | REVERSIBLE<br>IMMEDIATELY |
|-----------------|----------------------------------------------------------|-------------|---------------------------|

Three capabilities do not come home: the edge, outbound mail and volumetric scrubbing. Naming them now keeps the plan honest.

|        |                                             |                                                                                                                                                  |
|--------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | CloudFront and Shield                       | Shield Advanced bills \$3,000 a month on a twelve-month term and attaches only to AWS resources, so it protects a distribution and never a rack. |
| GOOGLE | Cloud CDN and Cloud Armor                   | Cloud CDN is a mode on a load balancer backend, so a colocated origin must first become an internet network endpoint group.                      |
| AZURE  | Front Door and its web application firewall | managed rule sets need the Premium tier, and the firewall policy is a resource that outlives its profile.                                        |

## BEFORE YOU START

### ACCESS

- The three invoice lines for edge, mail and mitigation, and authority to sign

### SOFTWARE

- The comparison from Move 03, open for a new permanent line
- The sending domain's current SPF, DKIM and DMARC records

### PEOPLE

- Whoever owns the marketing sends, who shares the reputation
- An engineer willing to name the fourth item honestly

## WHY THIS WORKS

A plan that pretends everything comes home fails in Stage 5, when somebody notices password resets landing in spam folders and no schedule is left. Three capabilities are bought rather than built, because building them badly is worse than renting them well: a hundred points of presence, a decade of sender reputation, and absorption capacity upstream of your transit. None is expensive next to what it prevents, and all three are worth paying for after the move on the terms you pay now. Deciding here costs two days and removes the largest late surprise.

### SWAP

One vendor covering delivery and mitigation halves the contracts, and one outage then takes both.

### DO IT FASTER

Ask each vendor for the contract and onboarding checklist in the first message.

### WATCH OUT

Facility-included protection is often a null route with a better name. Get in writing what it does to your address.

### LEFTOVERS

The cloud edge stays in front of the cloud origin until Move 18 shifts traffic, so two edge bills overlap.

|                   |                   |             |                  |                  |           |
|-------------------|-------------------|-------------|------------------|------------------|-----------|
| \$5,200/mo<br>WAS | \$5,200/mo<br>NOW | 0%<br>SAVED | 0 min<br>CUTOVER | 2 days<br>EFFORT | —<br>WAIT |
|-------------------|-------------------|-------------|------------------|------------------|-----------|

TURN OFF Nothing, and that is the point. The cloud's versions switch off in Moves 18 and 20; these three stay on the invoice permanently.

NEEDS 02 What you actually run 03 The number that decides it

## THE RUNBOOK

- 01 | Price the edge. Take the delivery and egress figures off the invoice from Move 01 and quote three independent vendors against that volume. A content network in front of a rack is the same arrangement as in front of the cloud.
- 02 | Price outbound mail on its own. Transactional sending goes to a provider whose only business is deliverability, and the sending domain's SPF, DKIM and DMARC records point at it. Marketing sends take a separate subdomain.
- 03 | Price mitigation upstream of your transit, the only place it works. Ask the facility and each transit vendor what is included, at what capacity and how it triggers: a hundred-gigabit attack against a ten-gigabit port is decided before it reaches your equipment.
- 04 | Sign all three contracts before the hardware ships, confirming each notice period and minimum term. Mail warm-up takes weeks and cannot begin during an incident.
- 05 | Write the total, about \$5,200 a month, into the comparison from Move 03 as a permanent line. Then name the fourth item: whatever the business depends on that nobody here has run, such as a payments integration.

## ROLLBACK

Nothing here touches a running system, so backing out is cancelling inside a notice period. The point of no return arrives later, when the sending domain's records point at the new provider: reputation then accrues on their addresses, and the old standing cannot be restored by changing the records back.







# II

# BUY

What to order and where to put it. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy. Both clocks start in this stage, and they run in parallel.

|       |               |           |                |
|-------|---------------|-----------|----------------|
| 4     | 4             | 2         | 0              |
| MOVES | ZERO DOWNTIME | HIGH RISK | MINUTES, TOTAL |

---

IF YOU ONLY DO THREE

05

## FROM RENTED VCPUS TO CORES YOU OWN

A cloud vCPU is a scheduling unit. Measure the workload before buying cores, then leave room for the guests and a ...

06

## SIXTEEN MACHINES, AND THE TWO ON THE SHELF

Sixteen active nodes and two reserved replacements, bought for the shelf or included in the rental quote, because ...

FROM RENTED VCPUS TO CORES YOU OWN · SIXTEEN MACHINES, AND THE TWO ON THE SHELF · A CAGE, NOT A DATA CENTRE · THE ORDER, AND THE WEEKS YOU CANNOT COMPRESS

---

■ 2 MEDIUM RISK

■ 2 HIGH RISK

---

# 05 FROM RENTED VCPUS TO CORES YOU OWN

CUTOVER

0 MIN



|              |                                           |                |                                         |
|--------------|-------------------------------------------|----------------|-----------------------------------------|
| LAYER<br>BUY | LEAVING<br>THE VCPU AS A UNIT OF PURCHASE | RISK<br>MEDIUM | REVERSIBLE<br>UNTIL THE ORDER IS SIGNED |
|--------------|-------------------------------------------|----------------|-----------------------------------------|

A cloud vCPU is a scheduling unit. Measure the workload before buying cores, then leave room for the guests and a failed host.

|        |                              |                                                                                                               |
|--------|------------------------------|---------------------------------------------------------------------------------------------------------------|
| AWS    | EC2 instance types           | a vCPU is one SMT thread on many x86 families but a whole physical core on Graviton, labelled the same.       |
| GOOGLE | Compute Engine machine types | custom shapes set memory apart from cores, within a per-core band that drags cores onto the bill.             |
| AZURE  | Virtual Machine sizes        | the B-series meters CPU against a credit balance, so a saturated workload reads low once its credits run out. |

## BEFORE YOU START

### ACCESS

- The two-week measurement window from Move 01, working set sampled apart from CPU
- The instance inventory from Move 02, exported with family and size

### SOFTWARE

- A sizing sheet other people can check, with the derate and headroom visible
- The vendor documentation for each instance family you run

### PEOPLE

- Whoever signs the order in Move 06, present for the derate rather than shown it

## WHY THIS WORKS

Move 01's samples become a part number here. On many x86 families two vCPUs share a physical core; that identifies the source topology, not the throughput of a replacement processor. Benchmark the workload on the proposed hardware. Existing VMs share a Proxmox VE host as KVM guests, each with its own operating system and resource allocation. You are buying a pool of capacity, not one physical machine per cloud instance. Memory, peak CPU demand and the capacity left after a host fails decide how many guests fit.

### SWAP

With the window unfinished, size against last quarter's billed peak and mark the sheet provisional. It lands a third high.

### DO IT FASTER

Most estates run five workload shapes wearing forty names. Cluster them and the conversion takes an afternoon.

### WATCH OUT

Rightsizing advice optimises for a smaller rented instance, a different question from how much silicon to own.

### LEFTOVERS

Local NVMe delivers the provisioned IOPS line as a property of the drive. Check it against your measured write rate.

|     |     |       |         |        |      |
|-----|-----|-------|---------|--------|------|
| —   | —   | —     | 0 min   | 3 days | —    |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** Nothing yet, though the re-measurement in step two usually finds a service two sizes above its demand — shrink it and the bill falls this month.

**NEEDS** 01 The bill, and the three lines that are most of it 03 The number that decides it

## THE RUNBOOK

- 01 Group the instance inventory by family and mark each thread-per-vCPU or core-per-vCPU against the vendor documentation. On Graviton and the other Arm shapes a vCPU is already a whole core, and halving buys double.
- 02 Pull the burstable families out of the measurement window. B-series and t3 in standard mode meter against a credit balance, and once the balance is empty a saturated workload pins at its baseline — twenty per cent on a t3.medium. t3 defaults to unlimited mode, where the number is honest and the burst above baseline arrives on the invoice instead. Re-measure either way before the figure reaches the sizing sheet.
- 03 Enter the source topology in the sizing sheet, then benchmark a representative workload on the proposed CPU. Record peak demand and latency at that load; halving a vCPU count is not a performance test. Rebuild Arm applications for x86 before testing them on this reference hardware; an Arm disk image will not boot as an x86 KVM guest.
- 04 For VMs, budget guest RAM plus Proxmox VE, QEMU and storage reserves; do not sell the same RAM twice through ballooning. For containers, use working sets plus kubelet reserve, page cache and eviction thresholds. In both cases prove that the surviving hosts carry peak load after the largest host is unavailable, without counting an unplugged spare.
- 05 Take clock over count on the latency path — most of a Postgres query plan executes on one core — then turn capacity and write rate into drives. Record guest vCPU allocations separately from physical cores, and hand the specification to whoever signs the order in Move 06.

## ROLLBACK

A specification is undone by writing a better one: change the derate and reissue it. Nothing has been bought. The point of no return is the signature in Move 06, after which a short memory line is a second purchase with a lead time. Keep the sheet in version control so a revision can be restored.

# 06 SIXTEEN MACHINES, AND THE TWO ON THE SHELF

CUTOVER

0 MIN



|       |                                               |      |                           |
|-------|-----------------------------------------------|------|---------------------------|
| LAYER | LEAVING                                       | RISK | REVERSIBLE                |
| BUY   | ELASTIC NODE CAPACITY AND CLUSTER AUTOSCALING | HIGH | UNTIL THE ORDER IS SIGNED |

Sixteen active nodes and two reserved replacements, bought for the shelf or included in the rental quote, because iron does not autoscale.

|        |                            |                                                                                |
|--------|----------------------------|--------------------------------------------------------------------------------|
| AWS    | EC2 Auto Scaling groups    | the fleet ceiling is a per-region vCPU quota, raised by support ticket.        |
| GOOGLE | Managed instance groups    | the instance template is immutable, so a shape change means a rolling replace. |
| AZURE  | Virtual Machine Scale Sets | the platform fault domain count is fixed at creation and cannot be edited.     |

## BEFORE YOU START

### ACCESS

- The per-node sizing from Move 05 and the service inventory from Move 02
- Authority to commit the purchase or rental budget chosen in Move 03

### SOFTWARE

- A quote comparison sheet with three vendors or rental providers priced line by line

### PEOPLE

- A vendor or provider contact who will quote complete machines and support
- Whoever signs the purchase or rental commitment, with the chosen route agreed before ordering

## WHY THIS WORKS

Fleet size is the quietest thing the bill decides. Three nodes are a quorum and nothing more: lose one and a third of the capacity goes with it. Sixteen and a dead machine costs six per cent, which is the difference between an incident and a note in the channel — and it is why a repatriation at this size is a far easier argument than the same one at a tenth of the bill. Buy enough usable capacity after control-plane reservations, storage overhead and a host failure; gross core count alone does not prove that the workloads fit. For owned colocation, two spare machines are built, burned in and configured with the active fleet, then unplugged and shelved. Remote hands can put one into service while a failed machine goes through repair or return. The rented route reserves the same two machines as spare capacity in its eighteen-server quote; agree the provider's replacement response and how that capacity is brought into service before ordering.

### SWAP

Lease the eighteen machines over three years. The total is higher, but the monthly line resembles the cloud bill it replaces.

### DO IT FASTER

Ask for a configuration the vendor stocks. Wanting 288 GB because the model said 240 turns a two-week delivery into an eight-week build.

### WATCH OUT

Rails, power leads, 25 GbE optics and the support contract are quoted separately. A bill of materials listing only servers is short by thousands.

### LEFTOVERS

A spare helps only if it stays current. Firmware, Proxmox and Talos versions drift. Update each spare on an isolated maintenance network, then return it to reserve; unplug owned spares. Enrol one in production only when replacing a failed host.

|     |     |       |         |        |         |
|-----|-----|-------|---------|--------|---------|
| —   | —   | —     | 0 min   | 3 days | 4 weeks |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT    |

**TURN OFF** Nothing for months. The cloud instances and managed clusters carry production until Move 18 shifts the traffic; this Move commits the selected purchase or rental budget.

**NEEDS** 02 What you actually run 05 From rented vCPUs to cores you own

## THE RUNBOOK

- 01 Put the per-node sizing from Move 05 beside the inventory from Move 02. Reserve capacity for host services and one failed physical machine before placing guests or containers. The reference fleet uses sixteen active nodes; the two spares are additional replacement capacity. Recalculate the workload budget for the chosen platform.
- 02 Write the specification on one page: dual-socket, 32 physical cores, 256 GB of ECC memory with every channel populated, four 3.84 TB mixed-use NVMe drives plus a mirrored boot pair, redundant power supplies, two 25 GbE ports, and a controller speaking Redfish. For Proxmox hosts, require hardware virtualisation and verify IOMMU support if assigning devices to guests.
- 03 Send that page to three vendors for colocation, or three dedicated-host providers for rented metal. Record complete capacity, support and delivery terms in the quote comparison sheet. For a purchase, mark which components may be refurbished and quote new drives and power supplies; for rental, confirm support for the selected host operating system, virtualisation and remote management.
- 04 Include two identical spare nodes alongside the sixteen active ones. For owned colocation, burn them in with the fleet, then unplug and shelve them for remote hands. For rented metal, reserve two servers and document activation while the provider repairs a fault. Include all eighteen in the price; exclude spares from running capacity, quorum and storage replicas.
- 05 Confirm the route chosen in Move 03 against the hardware and facility quotes or the rental quote before making a binding commitment. For owned colocation, have the capital approver sign the hardware purchase order and record the ship date. For rented metal, order the quoted fleet under the agreed support and notice terms; the provider supplies the facility, so no hardware purchase or separate cage order is needed.

## ROLLBACK

Before an order is accepted, both routes can be reversed by leaving the quote unsigned. The point of no return is the binding commitment: configured hardware may be non-returnable, while a rented fleet can carry minimum-term and notice charges. Check those terms before signing. An owned fleet can be resold at its then-current value; a rental ends under the provider's cancellation terms.

# 07 A CAGE, NOT A DATA CENTRE

CUTOVER

0 MIN



|       |                                 |      |                              |
|-------|---------------------------------|------|------------------------------|
| LAYER | LEAVING                         | RISK | REVERSIBLE                   |
| BUY   | THE REGION-AND-ZONE ABSTRACTION | HIGH | UNTIL THE CONTRACT IS SIGNED |

The facility supplies power, cooling, connectivity and remote hands. Your team operates the platform; rented metal includes the room and the machines.

|        |                                                      |                                                                                                                                            |
|--------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | Regions and Availability Zones                       | zone letters are shuffled per account, so one account's us-east-1a is another's us-east-1c, and cross-zone traffic bills both ways.        |
| GOOGLE | regions and zones                                    | a subnet covers the whole region rather than a zone, and quota is granted per region, so capacity in one zone is not capacity in the next. |
| AZURE  | regions, availability zones and zone-redundant tiers | zone support varies by region and by machine series, and a resource created without a zone cannot be moved into one later.                 |

## BEFORE YOU START

### ACCESS

- The committed power figure and machine count from Move 06
- Quotations from three facilities against one written specification

### SOFTWARE

- A comparison sheet with monthly charge, cross-connect price and notice period side by side

### PEOPLE

- A remote hands team with an agreed response time, plus two internal contacts authorised to direct work
- Whoever signs the term, before the price is agreed

## WHY THIS WORKS

Colocation puts the building, power, cooling and physical support under one facility contract. Remote hands can inspect equipment, replace labelled parts and move cables to an agreed runbook while your existing operations team manages the platform remotely. This Move adds a facility charge — about \$4,500 a month for space, transit, cross-connects and hands before usage-based power — in exchange for that support. Rented metal bundles the room and machines into the monthly server price. Price both routes before signing; either lets the team operate infrastructure without running a data centre.

## SWAP

Rented dedicated servers can shorten procurement and avoid a separate cage contract. Confirm delivery dates, private networking, address portability and replacement response before ordering. The reference budgets the same operations hours for both routes: remote hands covers contracted physical work in colocation, and the rental provider covers it on dedicated servers. Choose renting when keeping capital available matters more than owning the fleet.

## DO IT FASTER

Ask for the standard agreement and the cross-connect list before price is discussed; a day of reading decides which clauses are worth arguing.

## WATCH OUT

A second building doubles the hardware, the spares and the change surface, and is the commonest reason a repatriation stalls. One site with backups held elsewhere beats two nobody has exercised.

## LEFTOVERS

The cage invoices from handover while the cloud bill stays where it was; both run in parallel for about four months.

|     |            |       |         |        |         |
|-----|------------|-------|---------|--------|---------|
| —   | \$4,500/mo | —     | 0 min   | 4 days | 3 weeks |
| WAS | NOW        | SAVED | CUTOVER | EFFORT | WAIT    |

**TURN OFF** Nothing. This Move adds an outgoing instead of removing one; the cloud bill does not move until the first service does.

**NEEDS** 03 The number that decides it

## THE RUNBOOK

- 01 Write one specification — full rack, committed kilowatts, two feeds, cross-connects, remote hands, term — and send it to three facilities. Set the location around network latency, delivery access and the contracted physical response time.
- 02 Size the power commitment against what actually draws: sixteen active nodes at about 450 W draw 7.2 kW before switches. The reference starts with a 10 kW commitment; verify the switch load and the facility's continuous-load limit, with either feed able to carry the full rack if the other fails. Leave the two spares unplugged on the shelf.
- 03 Contract the remote hands scope: inspections, labelled disk and machine replacements, cabling and power actions under an approved runbook. Put ticket-to-touch, included hours, out-of-hours rates and a named escalation in writing. Give the team the rack map and spare-part locations; name two internal contacts who can authorise work remotely.
- 04 Ask which carriers have live fibre in the meet-me room and which of them will sell at your volume, then get the cross-connect price list in writing, installation and disconnection included. Three carriers and a cheap cross-connect beat better marketing.
- 05 Negotiate the exit with the entrance: a term no longer than the hardware life, a capped escalator, a notice period you could serve. Three years with twelve months' notice repeats the mistake you are here to correct.
- 06 Before anyone signs, price the alternative in the next column: the same eighteen machines rented by the month, with power, networking and hardware replacement terms specified. Compare current quotes at matching capacity and support. Rented metal removes the fleet purchase and the separate facility agreement; use Move 03's full-cost comparison to choose the route.

## ROLLBACK

Refusing a facility that will not put ticket-to-touch in writing costs nothing: before signature there is no position to unwind. The point of no return is the commencement date, after which departure runs on the notice period rather than on a decision. File the quotations that lost; at renewal they restore a price to argue from.

# 08 THE ORDER, AND THE WEEKS YOU CANNOT COMPRESS

CUTOVER

0 MIN



|              |                                                                  |                |                                            |
|--------------|------------------------------------------------------------------|----------------|--------------------------------------------|
| LAYER<br>BUY | LEAVING<br>PROVIDER-ASSIGNED<br>ADDRESSES AND MANAGED<br>TRANSIT | RISK<br>MEDIUM | REVERSIBLE<br>UNTIL THE ORDER IS<br>SIGNED |
|--------------|------------------------------------------------------------------|----------------|--------------------------------------------|

Everything with a lead time goes on one order this afternoon, so the queues run beside each other instead of one after another.

|        |                                    |                                                                                                                           |
|--------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| AWS    | Elastic IP and Direct Connect      | the cross-connect authorisation expires ninety days after issue, and an Elastic IP bills on after its instance is gone.   |
| GOOGLE | external IP and Cloud Interconnect | the partner variant hides the physical layer, so there is no cross-connect to order and capacity comes from a fixed list. |
| AZURE  | Public IP and ExpressRoute         | the circuit bills from creation, before the carrier has provisioned anything against its service key.                     |

## BEFORE YOU START

### ACCESS

- The signed facility agreement from Move 07 and its cross-connect price list
- Purchase authority to sign transit and circuits

### SOFTWARE

- The switch part numbers from Move 06, for the optic compatibility list
- A written order sheet, one promised date and one name per line

### PEOPLE

- A named technical contact the carriers can reach in week three

## WHY THIS WORKS

Every other stage of this migration is work you control. This one is queueing, and the only thing you control is when it starts. A cross-connect and an optic ordered in the same week arrive while the machines are being built; ordered in sequence they add a month each. So the whole list goes in on one afternoon, and you accept paying for two transit paths from the start, because one upstream is a single point of failure with a contract attached. Nothing is installed yet, so every line can be withdrawn up to signature.

### SWAP

Buy every line from the facility on one invoice. It costs more per megabit and takes four vendors off the critical path.

### DO IT FASTER

Ask for the authorisation templates before the agreement is countersigned; no queue starts until they are lodged.

### WATCH OUT

A cloud's interconnect paperwork expires if the cross-connect is not done inside ninety days, and it is re-issued rather than extended.

### LEFTOVERS

The blended transit stays on the invoice until your own circuit has carried a full month including a peak.

|     |     |       |         |        |         |
|-----|-----|-------|---------|--------|---------|
| —   | —   | —     | 0 min   | 3 days | 4 weeks |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT    |

**TURN OFF** Nothing yet — this Move only adds lines. The first it retires is the blended transit, once your own upstream has held a month.

**NEEDS** 07 A cage, not a data centre



## THE RUNBOOK

- 01 Order the cross-connects the day the facility agreement is countersigned: one per upstream, one for the out-of-band line, one spare. Ask whether the panels to the meet-me room are in place; that decides days against weeks.
- 02 Take the facility's blended transit for go-live, already in the building, and order a second upstream from a carrier on a different fibre route to run beside it. On the facility's own addresses that second circuit buys a price and a path to grow into rather than a failover, because only the facility's transit routes to you; it becomes a failover the day you announce your own prefix to both.
- 03 Take a /24 or a /25 and IPv6 from the facility. Your own autonomous system number can wait until a second site earns its eight weeks of registry paperwork. Rented space is a renumbering you may do twice.
- 04 Order the out-of-band line: a small separate circuit, or a mobile router. It is the cheapest item here and the one that matters at three in the morning.
- 05 Order optics and cables from the switch vendor's supported list, twenty per cent over the port count. Then chase the order sheet weekly.

## ROLLBACK

Until a signature this Move is undone by an email. The point of no return is the first signed term, not the first delivery: transit and cross-connects carry minimum terms in months, and a circuit against the wrong cabinet is paid for whether or not it is lit. Keep the quotes in version control, so a position can be restored at renewal.



# III BUILD

Everything between a racked machine and a cluster that could take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

|            |                    |                |                     |
|------------|--------------------|----------------|---------------------|
| 4<br>MOVES | 4<br>ZERO DOWNTIME | 3<br>HIGH RISK | 0<br>MINUTES, TOTAL |
|------------|--------------------|----------------|---------------------|

IF YOU ONLY DO THREE

09

**RACKING DAY**

Prove remote console access to the whole batch, then shelve the spares. Rails, power and labels take the day; the ...

12

**DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH**

Choose who owns each physical disk. Proxmox supplies VM datastores; Rook supplies storage on the direct Talos path. ...

10

**THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS**

Address ranges written down before anything is configured, two switches either of which may fail, and a way in that ...

RACKING DAY · THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS · THE PLATFORM YOUR WORKLOADS NEED · DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH

■ 1 MEDIUM RISK

■ 3 HIGH RISK

# 09 RACKING DAY

CUTOVER

0 MIN



|                |                                                                     |                |                           |
|----------------|---------------------------------------------------------------------|----------------|---------------------------|
| LAYER<br>BUILD | LEAVING<br>THE PROVIDER'S SERIAL<br>CONSOLE AND BOOT<br>DIAGNOSTICS | RISK<br>MEDIUM | REVERSIBLE<br>IMMEDIATELY |
|----------------|---------------------------------------------------------------------|----------------|---------------------------|

Prove remote console access to the whole batch, then shelve the spares. Rails, power and labels take the day; the proof takes twenty minutes.

|        |                                     |                                                                                                                                                                         |
|--------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | EC2 Serial Console                  | granted per account, Nitro types only, and no virtual media, so your own image never boots.                                                                             |
| GOOGLE | Compute Engine serial console       | switched on by the serial-port-enable metadata key, and an organisation policy can forbid it estate-wide.                                                               |
| AZURE  | Boot diagnostics and Serial Console | on by default against a managed storage account you cannot open, gated behind a role and killable subscription-wide, and reaches the guest only, never wedged firmware. |

## BEFORE YOU START

### ACCESS

- A booked day on site for two named people
- An out-of-band line that is live and independent of the uplinks

### SOFTWARE

- `ipmitool` and a Redfish client on a laptop outside the building
- A memory tester, a drive exerciser and one firmware bundle for the whole batch

### PEOPLE

- Two people for the lift; a chassis at head height is not a one-person job

## WHY THIS WORKS

The physical work is ordered so nothing has to be undone: rails across the active positions before a chassis is lifted, weight low, power split so one distribution unit cannot take the estate. None of that is the point. The Move exists for the baseboard management controller in each node — a small computer with its own processor, port and firmware, and the whole of the console the cloud gave you. Proved from outside the building, it turns two years of faults into an afternoon at a keyboard rather than a drive to the facility.

### SWAP

Remote hands will rack and cable to a drawing at about \$150 an hour; buy that for the lifting, not the controllers.

### DO IT FASTER

Address and password the controllers on a bench the week before, so the day itself is cabling and testing.

### WATCH OUT

Controllers answer on any interface they find; one reachable from the internet is a full-power console for a stranger.

### LEFTOVERS

Cardboard, the spare's rail kit, and the temporary address scope that hands out addresses for months.

|     |     |       |         |        |      |
|-----|-----|-------|---------|--------|------|
| —   | —   | —     | 0 min   | 6 days | —    |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT |

TURN OFF Nothing yet. The active fleet draws power in one building while the whole cloud bill runs in another.

NEEDS 06 Sixteen machines, and the two on the shelf 08 The order, and the weeks you cannot compress

## THE RUNBOOK

- 01 Fit the active fleet's rail pairs before any chassis goes in, checked against the cabinet's depth. Rack bottom-up, heaviest low, two people on each lift. Test the spares temporarily; their permanent home is the shelf.
- 02 A-side power from one distribution unit, B-side from the other, so a node's supplies never share a strip. Pull one feed per node and confirm it stays up.
- 03 Cable data in one colour and management in another, both ends labelled before anything is plugged in. Label the position, not the machine.
- 04 Give each controller a static address on the management VLAN, serial-over-LAN on, and a password the cluster cannot reach. Run each operation through `ipmitool` and Redfish, noting which answers.
- 05 Leave the building. From a laptop elsewhere, over the out-of-band line, watch a node POST, enter firmware setup, mount an image over virtual media and power-cycle it. Until that works from outside, nothing here is built.
- 06 Run the memory tester and the drive exerciser overnight on all eighteen nodes, level them all to one firmware bundle, and before leaving site take front and rear photographs, the port map and a file naming each active rack position and each spare. Unplug and shelve the spares after acceptance; they remain outside every production cluster.

## ROLLBACK

Nothing here carries data: a mistake is undone by unranking a machine the same afternoon. The point of no return is the day this cabinet takes production traffic, after which moving a cable is a change window. Firmware is the exception — a downgrade is sometimes refused, so record the shipped version per serial and keep the vendor's image so a working level can be restored.

# 10 THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS

CUTOVER

0 MIN



| LAYER | LEAVING                          | RISK | REVERSIBLE  |
|-------|----------------------------------|------|-------------|
| BUILD | CLOUD-MANAGED PRIVATE NETWORKING | HIGH | IMMEDIATELY |

Address ranges written down before anything is configured, two switches either of which may fail, and a way in that depends on neither.

|        |                                 |                                                                                                                                                                    |
|--------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | VPC and its subnets             | a subnet is pinned to one Availability Zone and its block is fixed at creation, and frames run at 9001 bytes inside the VPC but 1500 through the internet gateway. |
| GOOGLE | VPC networks and subnets        | the MTU belongs to the network rather than the instance, and every virtual machine must be restarted before it uses a changed value.                               |
| AZURE  | Virtual Network and its subnets | no broadcast or multicast on the wire, so gratuitous ARP and VRRP never worked and a floating address had to be a load balancer.                                   |

## BEFORE YOU START

### ACCESS

- Console access to both switches, and the out-of-band line from Move 09 answering
- The public address range from Move 08, and the private ranges the organisation already uses

### SOFTWARE

- `ipmitool` 1.8.19 or newer, to prove the baseboard controllers answer on the management range
- A packet test that sets the do-not-fragment bit, because a default-size ping proves nothing here

### PEOPLE

- Two engineers for the switch-pull drill, one at the rack and one watching the graphs

## WHY THIS WORKS

The address plan is cheap to get right today and expensive to change in a year, because by then it sits in a partner's firewall and on every machine. It is written first, on paper: separate ranges, each with twice the room the estate needs now. The switches come as a pair, since one switch is one maintenance window that takes everything down with it. MTU gets a step of its own, because a mismatch does not fail cleanly — it passes small requests and stalls large ones. Nothing here carries production traffic yet, which is what makes the week safe.

### SWAP

Where the switches will not do multi-chassis aggregation, give each node an address per uplink and let it route to both independently: more host configuration, fewer shared failure modes.

### DO IT FASTER

Generate both switch configurations from the address plan rather than typing them twice. Hand-built pairs differ in one port, and the drill is what finds it.

### WATCH OUT

The way back in must not depend on the fabric it exists to rescue. A jump host on the data VLAN, or one hosted only inside the Proxmox cluster it must rescue, cannot recover that cluster.

### LEFTOVERS

The cloud private network and its per-gigabyte charges keep running throughout, and nothing here removes them.

|     |     |       |         |        |      |
|-----|-----|-------|---------|--------|------|
| —   | —   | —     | 0 min   | 4 days | —    |
| WAS | NOW | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** Nothing yet. The cloud's private network carries every service you run until the traffic moves in Stage 5.

**NEEDS** 09 Racking day

## THE RUNBOOK

- 01 Write the address plan before touching a switch. Reserve management, storage, host and guest networks, and the public block from Move 08, none overlapping the office, the cloud accounts or anything a partner routes. For Proxmox, also reserve cluster communication and migration paths; storage recovery must not starve quorum traffic. Store the plan in source control.
- 02 Cable the two top-of-rack switches as a multi-chassis pair with a peer link, and run two uplinks from every node into different switches, bonded. One switch then fails as a component rather than as an outage.
- 03 Build the controller VLAN with no route to the internet. Reach baseboard controllers and Proxmox administration through the independent jump host, with access rules separating controllers from host management. Guest networks cannot reach either; host updates use controlled outbound access. Prove it from outside the building with an `ipmitool` power status on every node.
- 04 Set the MTU identically on every switch port, every bond, host bridge and guest interface on each path, and record the number beside the address plan. Undocumented mixed segments are how a fortnight disappears.
- 05 Test with full-size packets that refuse fragmentation, on every path that will carry traffic. A mismatch breaks about five per cent of it and reads as an application bug for a week.
- 06 Run the drill in daylight, with both engineers watching. Pull the power from one switch, confirm every bond keeps a member, then plug it back in and confirm the link rejoins. Nobody who declines this today will do it under load.

## ROLLBACK

Nothing here carries production traffic, so backing out is reverting one switch configuration and restoring the previous one from source control. The point of no return is a date rather than a command: once node addresses and partner firewall entries reference the plan from step one, renumbering becomes a weekend rather than an edit. Argue the ranges into shape now, not after Move 11.



# 11 THE PLATFORM YOUR WORKLOADS NEED

CUTOVER

0 MIN



|                |                                              |              |                           |
|----------------|----------------------------------------------|--------------|---------------------------|
| LAYER<br>BUILD | LEAVING<br>MANAGED VM AND KUBERNETES COMPUTE | RISK<br>HIGH | REVERSIBLE<br>IMMEDIATELY |
|----------------|----------------------------------------------|--------------|---------------------------|

Cloud VMs become KVM guests on Proxmox VE. Containers keep Kubernetes. Owning the hardware does not require changing the application's packaging.

|        |                          |                                                                                                                                      |
|--------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | EC2 and EKS              | instance-role credentials depend on metadata, while pod identity can depend on an IAM OIDC trust that your new cluster must replace. |
| GOOGLE | Compute Engine and GKE   | attached service-account credentials come from a metadata service; copying the boot disk does not bring that service with it.        |
| AZURE  | Virtual Machines and AKS | managed identities are Azure resources with role assignments; neither a guest image nor a pod manifest transfers that trust.         |

## BEFORE YOU START

### ACCESS

- The machines and VLANs from Move 10, with host roles assigned from Move 02's inventory
- An API virtual address for Kubernetes and management access restricted to the private VLAN

### SOFTWARE

- For VMs: the Proxmox VE 9.2-1 installer, its published checksum, and versioned guest OS templates with a matching QEMU guest agent
- For Kubernetes: a supported Talos release and compatible Kubernetes version pinned exactly in Git; talosctl matching Talos, and kubectl matching Kubernetes
- helm, the Cilium chart pinned to an exact version, and the cilium CLI

### PEOPLE

- A platform owner and someone to disconnect a host while the owner watches recovery

## WHY THIS WORKS

Bare metal is the hardware layer. For an estate with VMs, install Proxmox VE on the physical hosts and run those workloads in KVM guests. A guest keeps its operating system, service manager and deployment tooling; several guests share each host. Proxmox manages their lifecycle and placement. It does not recreate the cloud's metadata, identity or autoscaling services. For an estate already running containers, Talos Linux directly on the hosts remains a useful choice. In a mixed estate, Talos can also run in Proxmox guests, with physical failure domains and storage ownership made explicit. The strip below retains the container reference's control-plane charge. Add VM platform support, guest licences and operating time to Move 03 before using that model for a VM estate.

### SWAP

A container-only estate can skip Proxmox and boot Talos directly. Choose from the inventory, not from the name of the cloud service on the invoice.

### DO IT FASTER

Build a guest template once and clone it. Keep its image digest and configuration alongside the application release.

### WATCH OUT

Do not stack Rook OSDs on replicated virtual disks. Passing through physical storage restricts migration; Move 12 assigns each device one owner.

### LEFTOVERS

Schedule host and guest patching separately, and rehearse upgrades on the spare. A VM snapshot is not an independent backup.

|          |        |       |         |        |      |
|----------|--------|-------|---------|--------|------|
| \$220/mo | \$0/mo | 100%  | 0 min   | 4 days | —    |
| WAS      | NOW    | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** Nothing yet. Cloud instances and managed clusters still serve production. Their charges stop only after their workloads pass the later migration gates.

**NEEDS** 10 The network, and the way back in when it breaks



- 01 Write the host and guest plan in Git. Reserve CPU and RAM for the hypervisor, storage and a failed host as sized in Move 05. Enable hardware virtualisation in firmware for the VM pool. Keep the management and cluster-quorum networks away from guest traffic and storage saturation.
- 02 For that pool, verify the Proxmox VE 9.2-1 checksum and each empty boot pair's serial before the installer erases it. Apply security updates in the lab, record the tested package versions, then roll that set to the hosts. Create a Proxmox cluster and join the empty hosts; use at least three for quorum.
- 03 Create test KVM guests from the versioned OS templates on the local datastore reserved during host installation. Set CPU, RAM, boot mode, VirtIO disks and VLAN bridges; enable and test the QEMU guest agent. Pick a CPU model supported by every migration destination. Keep HA disabled: Move 12 moves ordinary guest disks to shared storage before production.
- 04 Where Kubernetes is needed, allocate three control-plane nodes on separate physical hosts, whether they are Talos guests or physical machines. For Talos guests, include the siderolabs/qemu-guest-agent extension in the pinned installer image. Distribute workers across hosts and reserve etcd memory and disk I/O. Guests on one host share its failure; keep capacity available for recovery.
- 05 Generate the pinned Kubernetes configuration with `talosctl gen config` and commit it. Confirm the installation disk on each node before applying it, because Talos repartitions that disk. Set the API endpoint to the virtual address, apply the configurations, then run `talosctl bootstrap` on one control-plane node and join the workers.
- 06 Install Cilium with `helm install` at the pinned chart version, with kube-proxy replacement enabled and the API host set to the virtual address. Disable kube-proxy in the same configuration change. Verify its absence with `kubectl`, then run the `cilium` connectivity test. Skip the Kubernetes steps for a VM-only estate.
- 07 Disconnect one physical host while only test workloads run. Kubernetes must retain API quorum and reschedule replaceable pods; its local guest disks remain unavailable with that host, so expect no VM failover yet. In Move 12, repeat with shared storage and VM HA, enforcing negative resource-affinity rules to keep control planes apart. A failed VM restarts; it does not live-migrate from a dead host.

## ROLLBACK

While only test guests and an empty cluster exist, restore the versioned configurations or rebuild from the recorded images. Reverting the kube-proxy setting restores that network path without changing the hosts. The point of no return is the first production disk or application write: after that, rebuilding a host requires preserving its datastore and restoring the affected workloads. Keep cloud production authoritative until the later Moves test migration and recovery.

# 12 DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH

CUTOVER

0 MIN



|                |                                                     |              |                           |
|----------------|-----------------------------------------------------|--------------|---------------------------|
| LAYER<br>BUILD | LEAVING<br>MANAGED BLOCK AND<br>SHARED-FILE STORAGE | RISK<br>HIGH | REVERSIBLE<br>IMMEDIATELY |
|----------------|-----------------------------------------------------|--------------|---------------------------|

Choose who owns each physical disk. Proxmox supplies VM datastores; Rook supplies storage on the direct Talos path. Replication belongs in one layer.

|        |                               |                                                                                                                                                         |
|--------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | EBS and EFS                   | a gp3 volume lives in one Availability Zone and its provisioned throughput is separate from capacity; EFS in bursting mode spends credits that run out. |
| GOOGLE | Persistent Disk and Filestore | Persistent Disk performance scales with provisioned size and the attached machine's vCPU count, and a Filestore instance grows but never shrinks.       |
| AZURE  | Managed Disks and Azure Files | Premium SSD steps in fixed tier sizes rather than a dial, and premium file shares need a FileStorage account that standard shares cannot move into.     |

## BEFORE YOU START

### ACCESS

- The device inventory from Move 09, with data drives identified by serial and boot drives excluded
- Proxmox storage administration or the Talos configuration from Move 11, according to the chosen platform

### SOFTWARE

- The Proxmox-supported Ceph package version pinned across hosts on the VM path
- helm and kubectl for the Talos path or external Ceph clients, with Rook 1.16.9 and a compatible Ceph image pinned in the deployment configuration
- ceph for health, capacity and recovery output during the drive-pull test

### PEOPLE

- One named owner for Ceph, including its object gateway and the recovery drill

## WHY THIS WORKS

A VM needs a datastore before it needs Kubernetes. On the Proxmox path, the hosts own Ceph and present shared block storage to guests. On the direct Talos path, Rook manages Ceph on physical disks. Running another replicated Ceph inside guests whose disks already sit on host Ceph multiplies copies and hides the physical failure domains. Choose one owner per device. PostgreSQL can use local NVMe because it replicates to its own standbys. That saves a second replication layer but binds each copy to its host. Its standbys must occupy different physical machines, even when the database runs inside Kubernetes VMs. The reference disk split and savings below describe the direct Talos build; price the VM datastore and its recovery capacity separately.

### SWAP

Proxmox can use existing shared storage or local datastores. Local disks need transfer during migration; host failure needs replication or restore, so demonstrate the promised recovery before choosing them.

### DO IT FASTER

Name storage classes and datastores for the recovery promise they make, so each workload gets a deliberate placement.

### WATCH OUT

Snapshotting a volume is not an independent backup. Keep a tested copy beyond the storage cluster that serves it.

### LEFTOVERS

The managed volumes, file shares and snapshots keep billing until their contents have moved and the retention window ends.

|            |          |       |         |        |      |
|------------|----------|-------|---------|--------|------|
| \$9,400/mo | \$850/mo | 91%   | 0 min   | 6 days | —    |
| WAS        | NOW      | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** Nothing this week. The managed disks and shares carry production until Stage 4 moves what is on them; the line dies in Move 15 for the buckets and Move 16 for the database.

**NEEDS** 11 The platform your workloads need

- 01 Record each disk's owner and confirm the selected devices are empty before provisioning. On direct Talos, reserve three data drives per participating node for Rook and the fourth for local PostgreSQL; exclude shelf spares. On Proxmox, assign host datastore disks separately from any local database devices.
- 02 For Proxmox, use its Ceph wizard with the pinned package version, monitors on separate physical hosts, and only the recorded datastore devices. Add an RBD pool as VM storage. For direct Talos, install Rook using `helm install rook-ceph rook-release/rook-ceph --version 1.16.9`, then apply the pinned CephCluster configuration selecting the inventoried raw disks.
- 03 Set replicated pools to three copies and `min_size 2`, with the failure domain at the physical host. Use `ceph df` and model redistribution after losing the largest storage host; keep utilisation below 70 per cent and verify recovery still has room. Guest count is not a count of independent replicas.
- 04 Provision the shared filesystem and object gateway required by Moves 15 and 16. Rook creates these on direct Talos. For host-owned Ceph, its owner provisions CephFS and a separate RADOS gateway service; an RBD datastore alone supplies neither an S3 endpoint nor its credentials. Connect Kubernetes through Rook's external-cluster mode, with restricted client keys and the existing gateway endpoint, never guest OSD disks.
- 05 Keep each local PostgreSQL copy on a different physical host, with placement rules enforced during failover. If passing a disk or controller into a guest, give it exclusive ownership and exclude it from host Ceph. That guest loses ordinary live migration; database replication and a tested restore provide its recovery path.
- 06 On Proxmox, move ordinary guest disks from Move 11's local bootstrap datastore to shared RBD. Verify every required disk and network exists on eligible hosts, then register those guests as HA resources with negative resource-affinity rules separating control planes. Exclude guests tied to local or passthrough devices. Write test data to a VM disk or a claim created with `kubectl` on Talos. Power off one physical host and verify recovery and contents, including the filesystem and gateway. Confirm no failure removes every database replica.
- 07 With only verified test data present, pull a storage drive and time recovery with `ceph -s`. Reseat it, wait for health to recover, and record the duration before accepting production data.

## ROLLBACK

Before test data lands, backing out means removing the empty pools and restoring the recorded disk layout. The point of no return is the first production write: leaving either Ceph or a local datastore then requires a data migration with a write stop. Back up a test VM or volume outside this cluster, restore it elsewhere, and verify its contents before acceptance. A local database copy needs its own restored backup as well as surviving replicas.



# IV MOVE

The application first, then the state. The stage that can lose a company its data, and therefore the one with the longest overlap windows, the most rehearsal and the fewest one-way steps.

|       |               |           |                |
|-------|---------------|-----------|----------------|
| 4     | 3             | 2         | 15             |
| MOVES | ZERO DOWNTIME | HIGH RISK | MINUTES, TOTAL |

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IF YOU ONLY DO THREE

13

## IMAGES, SECRETS AND ONE-COMMAND DEPLOYS

Three pieces of plumbing, then staging moves onto them: the first real workload on hardware you own, and the only one ...

15

## BUCKETS, CACHE AND QUEUES

Objects, cache and queues move by copying to a second place, then changing an endpoint. None of them needs writes ...

14

## THE FIRST SERVICE, END TO END

The first service runs in a VM or a container on your hardware, with its data still in the cloud and a traffic weight ...

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IMAGES, SECRETS AND ONE-COMMAND DEPLOYS · THE FIRST SERVICE, END TO END · BUCKETS, CACHE AND QUEUES · POSTGRES, THE ONE THAT MATTERS

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■ 2 MEDIUM RISK

■ 2 HIGH RISK

---

# 13 IMAGES, SECRETS AND ONE-COMMAND DEPLOYS

CUTOVER

0 MIN



|        |                                                               |          |             |
|--------|---------------------------------------------------------------|----------|-------------|
| LAYER  | LEAVING                                                       | RISK     | REVERSIBLE  |
| ➔ MOVE | MANAGED REGISTRIES,<br>SECRET STORES AND<br>HOSTED CI RUNNERS | ■ MEDIUM | IMMEDIATELY |

Three pieces of plumbing, then staging moves onto them: the first real workload on hardware you own, and the only one whose bad afternoon costs you nothing.

|        |                                      |                                                                                                                                                 |
|--------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | ECR and Secrets Manager              | the registry token expires every twelve hours, and a secret you remove holds its name for thirty days unless you ask for the seven-day minimum. |
| GOOGLE | Artifact Registry and Secret Manager | values are immutable, so every rotation leaves a billable version until destroyed.                                                              |
| AZURE  | Container Registry and Key Vault     | soft-delete cannot be switched off, and purge protection, once on, holds the name ninety days.                                                  |

## BEFORE YOU START

### ACCESS

- A Git repository the deployment tooling can read, with a deploy key held outside the platform
- Access to the existing image registry or VM artifact store and the Proxmox templates

### SOFTWARE

- For containers: Argo CD 3.0 and Harbor 2.12, installed with heLM at pinned chart versions, and an ephemeral runner controller
- For VMs: the existing versioned application deployment and runner tooling
- sops and age for encrypting secrets, with the private key kept outside the platform

### PEOPLE

- One engineer who owns the repository layout; two owners produce two conventions

## WHY THIS WORKS

Nothing real has moved yet, so this is the moment for plumbing. Container workloads use a registry and Kubernetes manifests reconciled by Argo CD. VM workloads keep versioned OS templates and their existing application deployment recipes in Git; the hypervisor does not replace that tooling. Secrets encrypted under a key held outside the platform let either kind of workload recover its credentials. The detailed registry and runner steps below are the container reference. A VM-only estate can retain its existing artifact store and runner service while proving the same deployment and recovery checks. Staging goes before customers do. Exercise the same VM templates or container images, secret delivery and deployment path production will use, against staging's own data. A guest that loses its configuration on reboot and a pod whose volume will not bind both belong here. Keep staging on the new platform for a fortnight of real branch deployments before Move 14 introduces production traffic.

## SWAP

A plain registry over the same object store drops the database and cache Harbor needs, and with them projects, quotas and scanning.

## DO IT FASTER

Do the cache in an afternoon and leave the registry of record a fortnight; most of the benefit arrives with the first half.

## WATCH OUT

The registry has to run for the cluster to start, and the cluster for the registry to serve. Power the rack down and up once, deliberately, before an outage asks it.

## WATCH OUT

Staging that still talks to production data stores has not moved, it has been relocated. Point it at its own copies before step seven.

## LEFTOVERS

The cloud registry bills storage and egress until its images go, a Move 20 decision, and hosted runner minutes continue for anything still routed at them.

|            |          |       |         |        |         |
|------------|----------|-------|---------|--------|---------|
| \$2,600/mo | \$260/mo | 90%   | 0 min   | 7 days | 2 weeks |
| WAS        | NOW      | SAVED | CUTOVER | EFFORT | WAIT    |

**TURN OFF** The hosted runner minutes, once a week of builds has passed on your own nodes with no fallback. The non-production estate's cloud compute, once the fortnight in step seven is done and nobody has asked to go back. The cloud registry and its secret store stay funded until Move 20.

**NEEDS** 11 The platform your workloads need 12 Disks: what goes local, what goes on Ceph

- 01 For container workloads, stand the registry up as a cache. `helm install` Harbor at chart version 1.16.2 on the Ceph object store from Move 12, proxying the registry you use today. Repoint pulls and confirm workloads still start. A VM-only estate skips the registry steps and retains its artifact store.
- 02 Make it where images are pushed. Move the pipeline's push target across and mirror ninety days of tags, so both registries hold the same images while yours is on trial.
- 03 For Kubernetes, `helm install` Argo CD at chart version 8.1.3 so the repository rebuilds namespaces, deployments, network policy and storage classes. For VMs, record template checksums, guest resource settings and the pinned application deployment recipe there instead. Bootstrap credentials stay outside the platform.
- 04 Encrypt every secret into the repository with `sops`, under an age key kept in a password manager and one offline copy. Feed decrypted secrets through the existing VM deployment tool or the Kubernetes bootstrap path. Limit the bootstrap key to that process; never bake it into a template or image.
- 05 After backing up staging data, rebuild a disposable VM from its template and deployment recipe, or delete a non-production namespace and let the reconciler recreate it. Check that the workload and its secrets return; restore the staging data and verify its contents.
- 06 For Kubernetes runners, install the ephemeral controller at chart version 0.12.1 and route builds to it by label. VM deployments can keep their current runners. Test both routes by changing one line: the pipeline must build the pinned artifact and deploy to the intended guest or namespace without manual repair.
- 07 Move staging and review environments to their assigned guests or namespaces, update DNS, and use them for a fortnight of branch deployments. Their data may remain in the cloud, but only in staging's own stores. Reboot a guest or redeploy a container for each route in use; neither passes until its configuration survives.

## ROLLBACK

Revert the deployment recipe and route builds back to the existing runners. Restore staging data before returning a workload that has accepted writes; a Git revert cannot recover those writes. The point of no return is removing the old artifacts and credentials before their replacements have passed a rebuild. Keep the cloud registry, VM artifacts and secret store available until the corresponding guest or cluster has recovered from its recorded configuration.



# 14 THE FIRST SERVICE, END TO END

CUTOVER

0 MIN



|       |                                |        |             |
|-------|--------------------------------|--------|-------------|
| LAYER | LEAVING                        | RISK   | REVERSIBLE  |
| MOVE  | CLOUD VM AND CONTAINER COMPUTE | MEDIUM | IMMEDIATELY |

The first service runs in a VM or a container on your hardware, with its data still in the cloud and a traffic weight as the way back.

|        |                                     |                                                                                                                                                              |
|--------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | EC2 and ECS on Fargate              | VM Import/Export excludes Marketplace images and images with encrypted EBS snapshots; its Windows and SQL Server image restrictions also block many exports. |
| GOOGLE | Compute Engine and Cloud Run        | custom images export through Cloud Storage, but Google-provided Windows Server licences do not permit running those images outside Google Cloud.             |
| AZURE  | Virtual Machines and Container Apps | direct VHD download cannot use a disk attached to a running VM; a snapshot can be exported instead, but live snapshots are only crash-consistent.            |

## BEFORE YOU START

### ACCESS

- Proxmox management access and the VM configuration repository, or write access to the Argo CD application repository
- Permission to edit weighted DNS records and the required cloud credentials

### SOFTWARE

- For VMs: the versioned OS template and existing application deployment tooling; qm for an optional disk import
- For containers: kubectl and argocd, pointed at the new cluster
- dig for checking public DNS during the ramp

### PEOPLE

- The service owner, with test requests, an abort threshold and any guest licence approval
- Whoever maintains partner egress allowlists, warned in advance

## WHY THIS WORKS

The first service produces evidence before it produces savings. Its database remains in the cloud, so the new variables are the host platform, network path and identity. For a VM workload, create a fresh KVM guest on Proxmox VE and deploy the same application using its existing tooling. No container conversion is required. For a container workload, deploy the existing image to Kubernetes. Both routes can run beside the cloud copy and accept the same traffic ramp. This Move covers stateless services, including those installed inside VMs. A writable application disk makes a VM stateful even when nobody calls it a database. Keep that data authoritative in the cloud, or defer that VM until its owner has rehearsed the data transfer, write stop and return path for Move 18. The zero-minute strip is for parallel stateless deployment, not a promise about exporting a running machine.

### SWAP

A proxy you already control can shift request weight faster than DNS. Keep the same acceptance and abort thresholds.

### DO IT FASTER

Reuse the verified guest template and deployment recipe for the next VM with the same operating system.

### WATCH OUT

Image export preserves bytes, not cloud identity, instance-store data or attached services. Verify every dependency from the inventory.

### LEFTOVERS

Exported images, temporary buckets, snapshots, SAS links and both allowlist entries need dated cleanup after verification.

|             |        |       |         |         |         |
|-------------|--------|-------|---------|---------|---------|
| \$30,980/mo | \$0/mo | 100%  | 0 min   | 20 days | 8 weeks |
| WAS         | NOW    | SAVED | CUTOVER | EFFORT  | WAIT    |

**TURN OFF** Each old VM or container deployment after its own completed soak and return-path check. Shared instance groups, clusters and log retention stay until their last dependent service has passed; retain the snapshots required by the rollback plan.

**NEEDS** 13 Images, secrets and one-command deploys



## THE RUNBOOK

- 01 Choose a low-impact stateless service from Move 02. Identify every disk write, scheduled job and metadata dependency. Disable duplicate jobs in the test copy, and keep it isolated from production traffic until the owner confirms it cannot create a second writer.
- 02 For a VM, create a KVM guest from the OS template and deploy the pinned application release with its existing tooling. For containers, commit the manifests and let `argocd` sync them. Keep the cloud database endpoint. Replace metadata credentials with an explicit workload identity or scoped credential, and update partner allowlists.
- 03 If rebuilding a guest is impractical, check the provider's export restrictions and licence portability first. Take an application-consistent copy of every required disk, export through the provider console, then use `qm disk import` to import a supported image. Attach the disks, match BIOS or UEFI and set boot order. Rehearse this off the live path; an export needing a source shutdown gets a separate downtime window.
- 04 Boot the VM on an isolated VLAN and verify VirtIO drivers, networking, DNS, clock and service startup; an imported disk is not evidence of a working application. For either route, send synthetic requests, check cloud API access without metadata, and compare error rates and latency percentiles with the cloud copy.
- 05 Ramp the weighted record: one per cent, ten, fifty, all of it. Hold each step a business day, check resolver answers with `dig` and measure the actual request split in service logs. Stop on the owner's error or latency threshold. DNS weight is approximate and cached clients will lag behind it.
- 06 Soak a week at full weight. Watch guest disks through Proxmox and application metrics, or node and pod health with `kubectl`. Exercise a service restart and check that configuration survives; scratch files must not become untracked application state.
- 07 Repeat across the stateless inventory, batched by owner. The reference fleet budgets twenty days of work across an eight-week cadence. Measure VM rebuild and export effort separately; do not squeeze stateful disk transfers into that allowance. Retain each cloud copy until its own soak and return-path checks pass.

## ROLLBACK

Set the new copy's traffic weight to zero, wait for cached clients to drain, and confirm requests return to the cloud. Keep that copy healthy through the soak. There is no point of no return for this stateless path because its authoritative data never moved. If the new guest has accepted unique persistent writes, stop: returning requires reverse transfer or a restore that preserves those writes before traffic changes. A stale cloud disk or pre-cutover snapshot cannot supply them. That VM needed the separate stateful migration plan.

# 15 BUCKETS, CACHE AND QUEUES

CUTOVER

0 MIN



| LAYER | LEAVING                                  | RISK | REVERSIBLE |
|-------|------------------------------------------|------|------------|
| MOVE  | MANAGED OBJECT STORAGE, REDIS AND QUEUES | HIGH | 30 DAYS    |

Objects, cache and queues move by copying to a second place, then changing an endpoint. None of them needs writes stopped.

|        |                                                     |                                                                                                                        |
|--------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| AWS    | S3, ElastiCache and SQS                             | SQS now carries a megabyte a message, so a large payload may be sitting inline rather than as a pointer into a bucket. |
| GOOGLE | Cloud Storage, Memorystore and Pub/Sub              | Pub/Sub acknowledges per subscription, so two subscriptions deliver everything twice.                                  |
| AZURE  | Blob Storage, Azure Cache for Redis and Service Bus | sessions and duplicate detection begin at the Standard tier; Basic has no topics.                                      |

## BEFORE YOU START

### ACCESS

- Read access to every bucket and the managed cache, and write on the Ceph gateway (Move 12)
- The queue inventory: visibility timeouts, dead-letter rules and message size ceilings

### SOFTWARE

- rsync 1.68 or newer, comparing checksums rather than size and time
- Valkey 8.0 with valkey-cli, and NATS JetStream 2.10 with the nats command line

### PEOPLE

- The owner of each key pattern and each queue consumer: both switches are code changes

## WHY THIS WORKS

The database is the hard migration; this one runs beside it. Objects, cache and queues can each be written to two places at once, so the work is a copy repeated until the delta is boring, then a configuration change. Writes never stop. The risk is not the bytes but the assumptions in the code: an object store that promised more than the standard did, a cache holding sessions nobody labelled as state, a queue whose visibility timeout the consumer extends. Reading source finds those. Watching a transfer does not.

| SWAP                                                                                         | DO IT FASTER                                                                                                         | WATCH OUT                                                                                                                | LEFTOVERS                                                                                                     |        |      |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|------|
| One small bucket read by one service needs no repeated passes. Copy it once, verify it once. | Start the bulk copy the week the cluster comes up. It is bandwidth, not attention, and the last delta takes an hour. | Presigned links issued before the endpoint changed resolve to the cloud until expiry; somebody always set one to a year. | Non-current object versions bill on after the live ones go, and cache nodes bill until the last client moves. |        |      |
| \$12,800/mo                                                                                  | \$1,150/mo                                                                                                           | 91%                                                                                                                      | 0 min                                                                                                         | 8 days | —    |
| WAS                                                                                          | NOW                                                                                                                  | SAVED                                                                                                                    | CUTOVER                                                                                                       | EFFORT | WAIT |

**TURN OFF** The cache nodes and the managed queues, after thirty days and a clean billing cycle. The object line shrinks rather than going: the archive stays.

**NEEDS** 12 Disks: what goes local, what goes on Ceph 14 The first service, end to end

## THE RUNBOOK

- 01 | Sort the object estate by size and last access. Hot buckets move; archive stays, because egress on extracting terabytes exceeds years of its storage fee.
- 02 | Copy the hot buckets with `rc1one` onto the Ceph gateway, repeating until a pass takes minutes. A multipart entity tag is a digest of digests: it compares only if part sizes match.
- 03 | Change the endpoint, then run a last `rc1one` pass for what was written during the swap. Read the client code for listing order, read-after-write on overwrites, part size.
- 04 | Scan the cache key space by pattern. Patterns with no expiry are sessions, counters and locks — state, not cache — and wait for Move 16. The owner confirms; the rest point at Valkey 8.0, checked with `valkey-cli`, and take a cold start in a quiet hour.
- 05 | Stand up NATS JetStream 2.10 beside the old queues. Make consumers idempotent, then have producers write to both for a day, checking each visibility timeout against a long job.
- 06 | Drain the old queues to zero, watch `nats` take the load, then stop the old consumers. Both copies are retained for thirty days; nothing is deleted.

## ROLLBACK

Nothing here is deleted, so backing out is three configuration changes: object endpoint, cache endpoint, broker. Objects written to Ceph afterwards are restored by copying the other way. The point of no return is the first message acknowledged only on the new broker.

# 16 POSTGRES, THE ONE THAT MATTERS

CUTOVER

15 MIN



|       |                    |      |            |
|-------|--------------------|------|------------|
| LAYER | LEAVING            | RISK | REVERSIBLE |
| MOVE  | MANAGED POSTGRESQL | HIGH | 7 DAYS     |

Fifteen minutes of stopped writes, at the end of a week in which the new cluster was caught up, backed up, and restored once on purpose.

|        |                                         |                                                                                                                                                                                                                      |
|--------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | RDS for PostgreSQL                      | the default parameter group cannot be edited, so <code>rds.logical_replication</code> needs a custom group and a reboot.                                                                                             |
| GOOGLE | Cloud SQL for PostgreSQL                | storage auto-grows to hold retained write-ahead log and never shrinks, so it bills at the high-water mark.                                                                                                           |
| AZURE  | Database for PostgreSQL flexible server | <code>CREATE EXTENSION</code> fails until the name is in the <code>azure.extensions</code> allowlist, and anything wanting shared memory needs <code>shared_preload_libraries</code> too, which does need a restart. |

## BEFORE YOU START

### ACCESS

- Logical decoding enabled on the managed instance, with the reboot taken
- Credentials for a bucket on the object gateway from Move 12, separate from the cluster's
- The Kubernetes database platform from Moves 11 and 12, including physical-host replica separation

### SOFTWARE

- CloudNativePG 1.25 or newer, failed over by hand at least once already
- The Barman Cloud plugin for CloudNativePG, pinned, because the operator writes `archive_command` itself and will not take one from you
- `psql` and `pg_dump` no older than the managed server

### PEOPLE

- The application owner, present for the window when writes stop

## WHY THIS WORKS

The application already runs on your hardware from Move 14, in a VM or a container. This Move migrates its managed PostgreSQL service to CloudNativePG on Kubernetes, using Move 12's local NVMe and object gateway. The primary and standbys must occupy different physical hosts even when their Kubernetes nodes are guests. The initial copy runs before the maintenance window; a rehearsed restore and final replication catch-up make the fifteen-minute reference possible. A VM-only estate may keep its managed database under Move 04. An existing PostgreSQL VM does not need converting to CloudNativePG to leave the cloud: keep its VM deployment and use a separately rehearsed database replication, failover and backup plan at Move 18. The operator-specific steps below apply only when the inventory selects the Kubernetes database route.

### SWAP

A reverse subscription can shorten rollback only if the managed service permits it and its replication loop prevention and sequence repair have been rehearsed. Keeping the old instance alone does not keep its data current.

### DO IT FASTER

Split the publication into several subscriptions grouped by table size, so the initial copy runs in parallel, not table by table.

### WATCH OUT

Replication carries rows only. Sequences, large objects, schema changes and tuned parameters stay behind, and a sequence left at one collides on the first insert.

### LEFTOVERS

The replication slot on the source outlives its subscription and retains write-ahead log for a reader that has gone.

|             |        |       |         |         |      |
|-------------|--------|-------|---------|---------|------|
| \$19,500/mo | \$0/mo | 100%  | 15 min  | 12 days | —    |
| WAS         | NOW    | SAVED | CUTOVER | EFFORT  | WAIT |

**TURN OFF** The managed instance, its read replicas and their snapshots, after the seven days and once an invoice shows the line gone.

**NEEDS** 12 Disks: what goes local, what goes on Ceph 13 Images, secrets and one-command deploys 15 Buckets, cache and queues

## THE RUNBOOK

- 01 List the extensions and roles on the managed instance with `psql`. What exists only on the provider's fork is replaced, re-expressed, or blocks the Move.
- 02 Copy the schema with `pg_dump --schema-only` onto a CloudNativePG cluster of one primary and two standbys on local NVMe, deferring the biggest indexes.
- 03 Create the logical replication publication and subscription and let it run for days. Read `pg_stat_subscription` each morning: a stalled subscriber fills the source's disk.
- 04 Declare the object store as the cluster's backup destination so the Barman Cloud plugin drives it, turn on continuous archiving, watch the first write-ahead log segments arrive in the bucket, and only then take the full backup.
- 05 Restore that backup into a scratch cluster, replay to a chosen timestamp, and count rows against production. Time it. The cutover waits until this passes.
- 06 Open the window. Stop every writer, including jobs inside VMs. Wait for replication lag to reach zero, compare the final data, advance sequences and switch every application connection string. Start writes only on the destination, then verify both VM and container clients use it.

## ROLLBACK

Until destination writes resume, return connection strings to the source and release its write pause. The point of no return for that quick return is the first destination commit. After it, stop writers again and use the rehearsed reverse replication or restore-and-replay procedure to carry those commits back; verify data and sequence positions before reopening the source. A source backup predating cutover cannot recover newer writes. Retain the managed instance and its backups for seven days, and take destination backups throughout that period.



# V RUN

The traffic, the pager and the last account. Every article about leaving the cloud stops before this stage, and every migration that regrets itself failed inside it.

|       |               |           |                |
|-------|---------------|-----------|----------------|
| 4     | 3             | 3         | 10             |
| MOVES | ZERO DOWNTIME | HIGH RISK | MINUTES, TOTAL |

---

IF YOU ONLY DO THREE

17

## THE FRONT DOOR

The new edge goes up beside the one carrying users, with its own address and certificates, proved from a host file ...

19

## BACKUPS YOU HAVE RESTORED, AND THE PAGER

VM backups, cluster manifests and a continuous database archive, copied beyond the rack — then the drill that ...

18

## GO-LIVE, AND HOW YOU ABORT

The weight moves in four steps and comes back in one, which is why the time-to-live drops a day early and the old ...

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THE FRONT DOOR · GO-LIVE, AND HOW YOU ABORT · BACKUPS YOU HAVE RESTORED, AND THE PAGER · CLOSING THE ACCOUNT

---

■ 1 MEDIUM RISK

■ 3 HIGH RISK

---

# 17 THE FRONT DOOR

CUTOVER

0 MIN



|              |                                                                     |                |                           |
|--------------|---------------------------------------------------------------------|----------------|---------------------------|
| LAYER<br>RUN | LEAVING<br>MANAGED LAYER-7<br>BALANCERS AND<br>CERTIFICATE SERVICES | RISK<br>MEDIUM | REVERSIBLE<br>IMMEDIATELY |
|--------------|---------------------------------------------------------------------|----------------|---------------------------|

The new edge goes up beside the one carrying users, with its own address and certificates, proved from a host file entry while real traffic goes elsewhere.

|        |                                                            |                                                                                                              |
|--------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| AWS    | Application Load Balancer and ACM                          | the validation record must stay in the zone forever, because every renewal re-checks it.                     |
| GOOGLE | external Application Load Balancer and Certificate Manager | URL map, target proxy and forwarding rule are separate objects, and the reserved address outlives all three. |
| AZURE  | Application Gateway and Key Vault certificates             | the gateway reads its certificate through a managed identity, and withdrawing that grant fails the gateway.  |

## BEFORE YOU START

### ACCESS

- The public address range from Move 08, with one address held back for the gateway
- Write access to the VM configuration repository or the Argo CD repository from Move 13
- DNS challenge credentials restricted to certificate validation records
- An export of every listener rule, redirect, rewrite and health check from the managed balancer

### SOFTWARE

- For VMs: a supported reverse proxy and ACME client, plus Keepalived, with exact package versions and configuration pinned in the guest template
- For Kubernetes: helm and cmctl, with the controller chart versions pinned below
- openssl, to inspect the certificate each listener actually serves

### PEOPLE

- Whoever owns the two most valuable transactions, to walk them end to end first

## WHY THIS WORKS

An edge cannot be half-tested once it is live, so this one is built beside the live one. For Kubernetes, Envoy Gateway terminates TLS and routes to services. For a VM estate, a pair of proxy guests routes directly to application guests; Kubernetes is unnecessary. Either edge takes an address from Move 08 and reaches the services already tested on your hardware. No user arrives, because no DNS record points at it. That exposes the hard part cheaply: the listener rules and rewrites accumulated in the managed balancer over three years, which nobody has a full list of. Store those rules in versioned proxy configuration or Gateway API objects. The figures below retain the reference Kubernetes edge; price the VM pair separately.

|                                                                                                             |                                                                                                                                                                                          |                                                                                                                                                                                                                                       |                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWAP<br>Under a dozen rules, skip the table and write the routes directly. It earns its keep around thirty. | DO IT FASTER<br>Stand the gateway up against one low-traffic service and keep that route as a permanent smoke test. It exercises address, certificate and route before the table exists. | WATCH OUT<br>A renewed certificate and a reloaded one are different events. Alert on the expiry of the certificate actually served by each proxy, and verify that the standby has the new certificate before it takes the virtual IP. | LEFTOVERS<br>The managed balancer charges per hour and per capacity unit until Move 18, and its validation records must stay in the zone or its certificates stop renewing. |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                   |                 |              |                  |                  |           |
|-------------------|-----------------|--------------|------------------|------------------|-----------|
| \$2,400/mo<br>WAS | \$320/mo<br>NOW | 87%<br>SAVED | 0 min<br>CUTOVER | 4 days<br>EFFORT | —<br>WAIT |
|-------------------|-----------------|--------------|------------------|------------------|-----------|

TURN OFF Nothing yet. The managed balancer, its listener rules and its certificate service can switch off after Move 18's retention window and a verified replacement.

NEEDS 14 The first service, end to end 16 Postgres, the one that matters



THE RUNBOOK

- 01
- For VMs, deploy the pinned proxy template to a pair of guests, with placement rules keeping them on different physical hosts. Reuse your team's supported proxy where available. For Kubernetes, install Envoy Gateway with `helm install eg oci://docker.io/envoyproxy/gateway-helm --version v1.4.2` and cert-manager at v1.17.2; commit values for Argo CD.
- 02
- Assign one virtual IP from Move 08. For VM proxies, configure Keepalived to track proxy health and transfer it between guests on the same layer-2 network; permit its peer traffic and address announcements through the host bridges. For Kubernetes, use the cluster network's layer-2 announcement. Power off the physical host serving the address and measure recovery, then test a failed proxy process separately.
- 03
- Use DNS validation so issuance works before traffic moves. On VMs, schedule the pinned ACME client and securely deploy renewed certificates to both proxies with a reload hook. On Kubernetes, issue through cert-manager and check `cmctl status certificate`. Rehearse renewal against the ACME staging service, then check production certificates with `openssl s_client -connect`, including each proxy after failover.
- 04
- Read the balancer export line by line into a table: one row per listener rule, path condition, redirect, header rewrite, sticky-session setting and health check. The rule nobody remembers is the one a customer depends on.
- 05
- Translate each row into the VM proxy configuration or a Kubernetes HTTPRoute, and write one request that proves it. Point VM backends at the application guests and enable health checks. Priority ordering and specificity ordering disagree, so a rule shadowed for two years comes back to life once translated faithfully.
- 06
- Point a laptop at the new address with a host file entry and walk sign-in, the two most valuable transactions, a large upload and any websocket. Cut nothing over; Move 18 moves the traffic.

ROLLBACK

Nothing here serves users, so backing out is removing the test gateway or proxy guests and their address assignment, once no host file entry still points at them. There is no point of no return in this Move, by design. Keep the definitions in the repository so the edge can be restored in an afternoon.

# 18 GO-LIVE, AND HOW YOU ABORT

CUTOVER

10 MIN



| LAYER                                                                                 | LEAVING                                    | RISK                                                                                   | REVERSIBLE |
|---------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------|------------|
|  RUN | WEIGHTED DNS AND THE PARALLEL MANAGED EDGE |  HIGH | 7 DAYS     |

The weight moves in four steps and comes back in one, which is why the time-to-live drops a day early and the old edge stays up a week.

|        |                               |                                                                                                            |
|--------|-------------------------------|------------------------------------------------------------------------------------------------------------|
| AWS    | Route 53 weighted records     | a group weighted entirely to zero answers with every record in it, so drain by raising the survivor.       |
| GOOGLE | Cloud DNS routing policies    | a policy cannot coexist with a plain record of the same name and type, so creating one replaces that set.  |
| AZURE  | Azure DNS and Traffic Manager | weighting lives in the profile, not the zone, so the apex needs an alias record and a second time-to-live. |

## BEFORE YOU START

### ACCESS

- Write access to the authoritative zone, and a second person who has it too
- The change window agreed with whoever answers the support line
- For a stateful VM, a restored backup and a tested transfer and reverse-transfer plan approved by its data owner

### SOFTWARE

- dig run from two networks you do not control, and a resolver you have never used
- argocd sync windows for Kubernetes, or the VM pipeline's deployment hold
- The application-native replication or backup tooling already tested for each stateful VM

### PEOPLE

- One named decider, and the abort criteria they read out rather than argue

## WHY THIS WORKS

The edge ramp changes where requests enter; it must not choose between competing copies of writable data. Stateless services already passed Move 14, and the reference database moved in Move 16. Both edges must reach the same authoritative workload and data before their weights can be interchangeable. Lower the time-to-live ahead of the day, check actual client behaviour, and let each soak span a traffic cycle. Stateful VMs deferred from Move 14 need a separate service window before this edge ramp. Seed their data, rehearse the final transfer and time the return path. The strip's ten minutes covers the reference edge change only. Record each VM's measured downtime separately; if a write stop exceeds an hour, split the migration by independently movable workload or retain that VM until a shorter method is proved. An image export does not acquire a zero-downtime guarantee by appearing earlier in the book.

### SWAP

Where a contract forbids a freeze, run the ramp in the customer's quiet hours in smaller increments across a fortnight.

### DO IT FASTER

Take the one per cent step a week early, on its own. It finds the certificate and header faults while there is time to fix them.

### WATCH OUT

Some corporate resolvers and mobile stacks ignore the time-to-live outright and reach the old address for days. Hence the week.

### LEFTOVERS

The lowered time-to-live is itself a leftover, multiplying query volume and query billing on a zone nobody reads.

|            |        |       |         |        |        |
|------------|--------|-------|---------|--------|--------|
| \$8,900/mo | \$0/mo | 100%  | 10 min  | 3 days | 1 week |
| WAS        | NOW    | SAVED | CUTOVER | EFFORT | WAIT   |

**TURN OFF** The weighted record and the managed edge behind it, seven days after full weight, and with them the egress line that was most of the \$8,900.

**NEEDS 17** The front door

## THE RUNBOOK

- 01 Lower the record's time-to-live to 60 in the authoritative zone a day ahead, then confirm it with `dig` on several resolvers, not the registrar's view.
- 02 Write the abort criteria as numbers: error rate above a figure, tail latency above a figure, any data-integrity report at all. Name the decider on it.
- 03 Tell the clients a record cannot move: mobile builds with a pinned host, batch jobs holding an old address, third parties allowlisting one.
- 04 For each deferred stateful VM, seed a destination guest while the source remains authoritative, using its tested application-native replication or backup tooling. Restore into isolation, verify record counts or file checksums, and rehearse final transfer and reverse transfer. Book its measured write-stop window separately from the edge change.
- 05 In that VM's window, stop all source writers and timers; keep destination writers stopped too. Drain replication or transfer the final delta, validate the data, then switch traffic and start only destination writers. Retain the source stopped. If verification misses the deadline before destination writes, abandon the cutover and resume the source. Never split live writes between independent VM disks.
- 06 Before the edge ramp, verify both edges route to that same authoritative service and data. Keep destination backups running. If the old edge cannot reach the new authority, it is not an available traffic-only rollback; repair that path or retain the existing edge until it can.
- 07 Open the edge window and freeze releases with an `argocd sync` window or the VM pipeline's deployment hold. A release landing mid-shift cannot be told apart from the shift.
- 08 Move the weight to one per cent and soak. Then ten, then fifty, then all of it. Confirm actual traffic, errors and latency after each change. The ten minutes in the strip is the final reference step, not the earlier VM windows or the soak.
- 09 Keep the managed edge configured, resolvable and at zero traffic weight for seven days, with the new edge's weight positive. Verify the return route during the week. Nothing is deleted before that check passes.

## ROLLBACK

For an edge failure, restore the old edge's traffic weight and confirm requests reach the same authoritative service; cached clients may take longer than one TTL. For a stateful VM, the point of no return for a quick switch back is its first destination write. Stop writers, reverse-transfer those changes or restore and replay them to the source, validate the recovered data, then restart source writers and change traffic. Never boot a stale source as a writable fallback. The managed edge stays available for seven days, but its existence cannot replace that data recovery procedure.

# 19 BACKUPS YOU HAVE RESTORED, AND THE PAGER

CUTOVER

0 MIN



|              |                                                               |              |                           |
|--------------|---------------------------------------------------------------|--------------|---------------------------|
| LAYER<br>RUN | LEAVING<br>MANAGED BACKUP, ALARMS AND A RENTED PAGING SERVICE | RISK<br>HIGH | REVERSIBLE<br>IMMEDIATELY |
|--------------|---------------------------------------------------------------|--------------|---------------------------|

VM backups, cluster manifests and a continuous database archive, copied beyond the rack — then the drill that measures recovery time.

|        |                                            |                                                                                                                                                 |
|--------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | AWS Backup and CloudWatch alarms           | a vault under compliance-mode lock cannot be shortened or removed until its retention expires.                                                  |
| GOOGLE | Backup and DR Service and Cloud Monitoring | a backup vault's enforced retention makes a backup indelible until it expires, by you or by Google, so the copy outlives the decision to leave. |
| AZURE  | Azure Backup and Azure Monitor alerts      | a Recovery Services vault protects only resources in its own region, and soft delete holds removed items for fourteen days.                     |

## BEFORE YOU START

### ACCESS

- Object storage with object lock at an independent provider
- Five engineer-days reserved for the rebuild drill

### SOFTWARE

- vzdump, qmrestore and guest agents from Move 11's pinned templates
- veLero 1.15 and Move 16's Barman Cloud plugin, pinned in Git
- rc1one 1.68 for the copy out to the third provider
- Prometheus and Loki with a series budget agreed before the first scrape

### PEOPLE

- The existing operations rota, with a second escalation contact
- Provider support contacts and approved physical-intervention runbooks

## WHY THIS WORKS

The deliverable is the drill: rebuild from the repository and off-site backups while production stays untouched, with a stopwatch running. Compare measured recovery time with the agreed objective; a slow restore means the target is still unmet. Monitoring runs on your platform; alerting runs outside it, because a failed cluster cannot report its own failure. The existing cloud operations rota transitions with the services. Physical faults go to the contracted remote hands team or rental provider; your engineers retain software recovery and incident command within the same recurring-hours budget, validated during the pilot.

### SWAP

If the drill on the spare machine is disruptive, rent a dedicated server by the month and rebuild into that instead.

### DO IT FASTER

Make the drill a script, not a document. The second run then takes an afternoon rather than a week.

### WATCH OUT

A booted VM or healthy pod proves little about the data. Check volumes, database rows and application behaviour, and test a full host failure separately from the spare-machine restore.

### LEFTOVERS

Backup vaults bill after the resources they protected are gone, and one under a retention lock stays until it expires.

|            |            |       |         |        |      |
|------------|------------|-------|---------|--------|------|
| \$4,800/mo | \$1,400/mo | 71%   | 0 min   | 5 days | —    |
| WAS        | NOW        | SAVED | CUTOVER | EFFORT | WAIT |

**TURN OFF** Managed backup vaults, their cross-region copies, the alarms and their ingestion, and the rented pager seats — once a restore of your own has been timed and one real page has woken somebody.

**NEEDS** 16 Postgres, the one that matters 18 Go-live, and how you abort

## THE RUNBOOK

- 01 Schedule `vzdump` for Proxmox guests to archive-file storage outside their datastore, including every required virtual disk and VM configuration. Use the guest agent to coordinate filesystems and application-native backups for database consistency. Save host configuration and recovery credentials separately; passthrough devices need their own data backups. For Kubernetes, schedule `velero` for objects and volume data, and continuous database archiving through the Barman Cloud plugin.
- 02 Copy completed VM backup archives and Kubernetes backup data out with `rc1one` to a provider that is neither the rack nor the account being closed. Use object lock and separate credentials; the source cluster cannot remove retained copies. Verify that decryption keys survive its loss.
- 03 Temporarily activate a shelf spare or reserved rental server on an isolated network. Rebuild Proxmox and restore a representative VM with `qmrestore`, or rebuild a test Kubernetes cluster from Git and restore its volumes and database. Validate application data with outbound jobs disabled, then compare elapsed time and data loss against the recovery objectives. After the drill, return it to reserve and unplug an owned spare.
- 04 Stand Prometheus and Loki up on the active nodes against the series budget, with drop rules at scrape time. Include Proxmox host quorum, VM backup failures, storage health and free capacity, as well as application signals.
- 05 Put the alerting where the cluster cannot take it down. OneUptime runs the probes, the rota, the escalation and the status page from outside; the author of this book founded OneUptime, and the copyright page says so. Add a dead-man's-switch heartbeat.
- 06 Transition the existing cloud rota to the new platform. Route physical faults to remote hands or the rental provider under the support contract; keep software recovery with the operations team. Test both escalation paths, name a backup contact, record what waits until morning and measure the hours against Move 03's budget.

## ROLLBACK

Nothing here alters what serves traffic, so backing out is disabling the new backup schedules and leaving the managed alarms enabled. The point of no return is the day the provider's vaults age out: after that the only copy of last month is the one you wrote. Keep them until a restore from your own has been timed.



RUN · 0 MIN · HIGH RISK

Managed backup vaults, their cross-region copies, the alarms and their ingestion, and the rented pager seats — once a restore of your own has been timed and one real page has woken somebody.

# 20 CLOSING THE ACCOUNT

CUTOVER

0 MIN



|                                                                                       |                                                                 |                                                                                        |            |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|------------|
| LAYER                                                                                 | LEAVING                                                         | RISK                                                                                   | REVERSIBLE |
|  RUN | THE PROVIDER ORGANISATION, ITS AUDIT TRAIL AND ITS SUPPORT PLAN |  HIGH | 30 DAYS    |

The estate moved a month ago. The account did not: a dormant organisation still bills, still holds credentials, and still lets someone start an instance in it.

|        |                                             |                                                                                                                                                        |
|--------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| AWS    | the management account and its support plan | closing a member account takes the management account, but making one standalone first needs its own contact details, payment method and support plan. |
| GOOGLE | organisation and its projects               | project deletion is scheduled rather than immediate, and a lien refuses the request outright.                                                          |
| AZURE  | subscriptions and the Entra tenant          | the tenant refuses deletion while any subscription or application registration still exists.                                                           |

## BEFORE YOU START

### ACCESS

- Root credentials for the management account, held by two named people
- The registrar login, often held by one person who has since left

### SOFTWARE

- aws, gcLOUD or az signed in with an identity that can read billing
- rcLone 1.68 or newer to copy the audit trail and billing history somewhere you own

### PEOPLE

- Whoever answers the auditor, because the attestations you inherited become yours to produce
- The finance owner who has to see the invoice reach zero and say so

## WHY THIS WORKS

A migration that stops at the traffic cutover leaves about \$1,240 a month and a quieter second problem: live credentials, an audit surface someone will still ask about, and enough idle capacity to make drifting back the cheapest decision in the room. Nothing here is deleted until a full billing month has passed with the line at zero and no ticket filed. The failures that follow a migration are monthly rather than daily — the job that runs on the twenty-eighth and cannot find its bucket. Switching off and closing down stay four weeks apart, with everything irreversible on the far side of that gap.

### SWAP

Where one thing genuinely has to stay — an archive whose egress you will not pay — move it into a small account of its own and close everything around it.

### DO IT FASTER

Begin the key management waiting periods on the first day of the wait, not the last. Seven to thirty days is the AWS range; Google's can be set as long as a hundred and twenty, and a Key Vault holds keys for a retention period fixed when the vault was created. Nothing else waits on any of them.

### WATCH OUT

Committed spend does not end when usage does. A Savings Plans commitment keeps billing against the organisation after every resource inside it has gone.

### LEFTOVERS

Billing exports, budget alerts and marketplace subscriptions outlive what they were attached to. So does the support plan, which renews on its own anniversary rather than yours.

|            |        |       |         |        |         |
|------------|--------|-------|---------|--------|---------|
| \$3,200/mo | \$0/mo | 100%  | 0 min   | 4 days | 4 weeks |
| WAS        | NOW    | SAVED | CUTOVER | EFFORT | WAIT    |

**TURN OFF** The account itself, and with it the support plan, the idle gateways, the log ingestion and the snapshot retention still billing a month after the last user request left. What is left is a platform rather than a project.

**NEEDS** 18 Go-live, and how you abort 19 Backups you have restored, and the pager

## THE RUNBOOK

- 01 Take the final invoice apart with `aws`, `gcloud` or `az`. Expect address translation gateways nothing routes through, CloudWatch log groups still ingesting, snapshots on a retention policy nobody reviews, a support plan priced against spend that has gone, and two subscriptions from a proof of concept three years ago.
- 02 Extract what cannot be recovered later: the audit trail for the period your auditor requires, the billing history, and the compliance attestations you inherited. Copy them out with `rc1one`, then open one of each and check it reads.
- 03 Revisit the archive you decided in Move 15 to leave behind. Data parked in an account you are closing is not archived anywhere. Move it out, or agree in writing to delete it once someone has confirmed nothing still reads from it.
- 04 Move the domain before anything touches identity. Where the registrar sits with the provider — Route 53 is its own, and Azure fronts a partner under its own name — transfer it away, then put the lock back on.
- 05 Turn everything off and remove nothing. Stop the instances, drop the support plan to the free tier, keep the snapshots, and wait one full billing month with the compute lines at zero and no ticket filed. The snapshots and the volumes under the stopped instances go on billing, and are meant to: that residue is what keeps the month reversible.
- 06 Close it from the leaves inward. Federation and human users first, then access keys and roles, then the keys in Secrets Manager or Key Vault, because those decrypt archives you may still hold. Member accounts and subscriptions next; the organisation itself last.

## ROLLBACK

Until the closure request this is ordinary work: an account switched off can be switched back on in an afternoon, which is why the wait is four weeks and not four days. The point of no return is the request itself. AWS will reopen a closed account for ninety days, a cancelled Azure subscription keeps its data for thirty to ninety, and a Google project stays scheduled for deletion for thirty — so a restore of anything left inside is possible for thirty days on all three and impossible after ninety. Use the wait to confirm the audit trail and billing history read correctly from your own copies.



RUN · 0 MIN · HIGH RISK

The account itself, and with it the support plan, the idle gateways, the log ingestion and the snapshot retention still billing a month after the last user request left. What is left is a platform rather than a project.



# A FEW THINGS WORTH KNOWING

None of these is a Move. They are the ten things that make the other 20 work, and most of them are the things people learn in the second year rather than the first.

## THE BILL LAGS THE CHANGE BY A MONTH

A resource stopped on the third still bills for the first two days, and a reservation you no longer use bills for a year. Judge a Move by the invoice after next, never by the console.

## TWO PEOPLE KNOW, OR NOBODY DOES

The single most common failure mode of a self-run platform is not hardware. It is one engineer who understands the network, and a holiday.

## KEEP ONE FOOT IN THE CLOUD ON PURPOSE

An account with a working credential, a Terraform state and a warm standby is not a failure of nerve. It is the cheapest disaster recovery site you will ever have.

## WRITE THE RUNBOOK BEFORE YOU NEED IT

Every Move in this book is a runbook because that is the artefact that survives the person who wrote it. A migration that leaves no runbooks behind has to be done again by whoever comes next.

## CAPACITY YOU OWN IS CAPACITY YOU HAVE

The cloud taught a generation to treat headroom as waste. On your own metal, idle capacity is the thing that absorbs a bad deploy at four in the afternoon. Buy thirty per cent more than the model says.

## HARDWARE FAILS PREDICTABLY AND BORINGLY

Disks, fans, PSUs, optics, in that order, at a rate you can put in a spreadsheet. It is not the drama people expect. The drama is in the change you made on Friday.

## THE SAVING IS REAL; THE FREEDOM IS THE POINT

Most teams start this for the bill and finish it for the other thing: being able to answer a question about your own system without opening a support case.

## YOU CAN GO BACK

The industry talks about repatriation as though it were one-way. It is not. 0 of the 20 Moves here are genuinely irreversible and the other 20 are not, and knowing which is which is most of the skill.

The best infrastructure decision you make this year will probably be unglamorous, take a fortnight, save a five-figure sum and never be written up anywhere. That is rather the point.

### THE BOOK

20 Moves across five stages. 18 cost no downtime at all, 0 cannot be undone, and the whole book executed end to end costs 25 minutes of user-visible outage.

### THE TYPE

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are open source. Every number in this book is set in the mono.

### THE NUMBERS

Every price is a public list rate observed while writing, before any discount, and every saving is illustrative of a shape rather than a quotation. Check the arithmetic against your own invoice.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Move would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.